

TOPAZ: Noise-Aware Quantum Circuit Architectures via Geometric Wells in the Parameter Landscape

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Circuit design is usually an exercise in keeping gates unitary; deleting unused gates is a separate, discrete decision. TOPAZ (Topology OPTimization Ansatz with Zoning) instead treats topology—which entangling edges are present—as a continuous design variable: each nearest-neighbour pair carries a sum of three Pauli rotations (the LPTE), its unitary envelope recovered by polar projection, with a dual-MMA optimizer driving edge intensities and timings concurrently. Our central structural result is a topological-sparsification theorem (Theorem 1): if an entangling pair’s block carries no component along the ideal rotation, its primary gradient vanishes, so its intensities stop responding to the objective and the edge becomes removable. We are explicit about the scale on which that removal is tested—the pruning rule thresholds the post-softmax pair density, not the unnormalized parameter the theorem refers to—and we measure the rule’s behaviour rather than assuming it. We further measure the finite-depth closure of the chained algebra through $N = 8$: the $3(N - 1)$ per-pair $\{XX, YY, ZZ\}$ generators close to a Lie algebra of dimension $\dim \mathfrak{l} = 4^{N-1} - 1$ (odd N) or $4^{N-1} - 4$ (even N), one quarter of $\mathfrak{su}(2^N)$ up to parity constants. Methodologically, an equalized-budget re-evaluation of the H_4 VQE benchmark reverses the apparent ranking of structured versus hardware-efficient ansätze; the durable advantage is mitigation compatibility—ZNE/CDR post-processing remains physically valid on the shallow topology without clamping, whereas deeper forms violate physical bounds on the majority of seeds. We additionally report where the intensity-threshold pruning rule does *not* fire: under the global softmax normalization of Eq. (1) the per-pair density is bounded below by the optimizer’s own weight sharing, so at the operating point $\tau_\kappa = 0.05$ no edge of the 8-qubit chain is pruned in either seed tested, and a measured τ_κ sweep locates the threshold at which pruning activates and the fidelity cost at which it becomes destructive. The framework’s fixed points, structural wells—the sublevel-set basins of Definition 1—are reproducible across seeds and transfer across synthetic noise models.

Keywords: TOPAZ, topology optimization, quantum error mitigation, noise-aware circuits, parametric Trotter expression, structural well

1. Introduction

A quantum computation is a sequence of gates, and on near-term hardware every gate is an opportunity for error. Depolarizing and damping channels erode the state, readout corrupts the output statistics, and coherence windows bound the total circuit time; variational algorithms must therefore execute complex operations within strict depth budgets [1, 2]. The conventional response is not to redesign the circuit but to keep it and edit the measurement statistics afterwards. Zero-noise extrapolation [3], Clifford data regression [4], and virtual distillation [5] all act after the gates have run—they stretch, fit, or distill. None of them alters which entangling edges are present in the compiled circuit, and a near-zero intensity that nobody deleted still becomes a physical CNOT.

We approach the problem from the other direction: design the circuit before it executes rather than correcting its output afterwards. TOPAZ relaxes the unitarity constraint during training and exposes each edge’s intensity $\kappa(p)$ as a continuous design variable, so that the optimizer can be asked—in principle—to suppress an edge below a registered threshold and delete it before compilation. Whether that request is honoured is an empirical question, and we answer it directly in §6.6: at the operating point studied here it is not, for a structural reason we state there. The engine is the *local parametric Trotter expression* (LPTE; §3): an N -qubit linear chain in which each nearest-neighbour pair $(i, i+1)$ carries a sum of

three Pauli-rotation terms

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-iP \delta \tau_{i,P}}, \quad (1)$$

with intensities $r_{i,P} \in [0, 1]$ and timings $\tau_{i,P} \in [0, \tau_{\max}]$. The full weight vector over all $3(N - 1)$ pair terms is normalized once, globally, via softmax: the optimizer works with unconstrained weights $\tilde{\rho}$ and sets $r_j = \exp(\tilde{\rho}_j) / \sum_{j'} \exp(\tilde{\rho}_{j'})$, ensuring $\sum_j r_j = 1$ without projection. The per-pair density $\kappa(p) = \sum_P r_{p,P} \in [0, 1]$ is then the pair’s share of that global weight, not an independent per-pair normalization. Each block is a non-unitary operator; it is projected to its nearest unitary by the polar factor

$$\mathcal{P}_U(A) = A (A^\dagger A)^{-1/2}, \quad (2)$$

and the circuit is the product of the per-pair unitaries. Because the relaxation is continuous, gradients can flow while an edge is being removed, something a strictly unitary ansatz cannot do.

This has the shape of a classical structural topology-optimization problem [6, 7]: a discretized solid, a density variable on every element, a compliance residual, and an optimizer that removes material the residual does not need. TOPAZ draws direction from that field. The density on edge p is the total intensity $\kappa(p) = \sum_P r_{p,P}$; the residual is the fidelity under a Kraus noise model; and the optimizer is the method of moving

asymptotes, applied twice—once for intensities ρ , once for timings τ . The rule is that edges whose intensity falls below a registered threshold $\tau_\kappa = 0.05$ are pruned, as low-density elements are removed in the solid isotropic material with penalization (SIMP) method of structural optimization [6]. We emphasize that the debt to SIMP is one of inspiration, not of method: TOPAZ prunes by an explicit threshold and applies neither SIMP’s power-law penalization of intermediate densities nor its compliance objective. The continuous-relaxation-and-prune strategy is the common thread; the machinery is TOPAZ’s own. The continuous relaxation is load-bearing in our measurements; the discrete prune step is a thresholded rule that, as §6.6 documents, does not fire under the normalization actually used here.

TOPAZ has three mechanisms. (1) The LPTE sum with polar retraction provides a continuous relaxation of the unitarity constraint, allowing gradients to flow around barriers that would block a strictly unitary optimizer. (2) Parameter zoning treats ρ and τ as heterogeneous design variables updated by separate dual-MMA solvers with independent asymptotes and move limits; a discrete stage then removes any edge whose intensity falls below the registered threshold. (3) The optimizer consistently reaches a *structural well*—a basin in the noise-deformed parameter landscape where the noisy output is robust and, empirically, stable across seeds, noise models, and hardware. Of these, (1) and (3) carry the measured results reported here; (2)’s discrete stage is characterized in §6.6 and shown to be inactive at the stated operating point.

We demonstrate the framework on the 8-qubit FakeFaz simulator and in a scoped 8-qubit experiment on a single quietest path of the 156-qubit `ibm_fez` (Heron r2) processor. The compiled ansatz carries 18.9 ± 1.7 two-qubit gates across the ten-seed ensemble, matching the 18 reported for the submitted instance. We note explicitly that the intensity-threshold pruning step of §5.5.2 is *dormant* at the operating point studied here: under the global normalization of Eq. (1) the per-pair density does not fall below $\tau_\kappa = 0.05$ for any edge of the 8-qubit chain, and the measured gate count reflects no pruning-induced reduction (a τ_κ sweep in §6.6 locates the threshold at which the rule activates). Post-processing ZNE on top of TOPAZ adds a further DM gain (Table XI). On hardware, the submitted circuit uses 19 two-qubit gates at depth 76, versus 21 two-qubit gates at depth 78 for the single-step product-Trotter baseline, and its noiseless one-way state-preparation fidelity is 56.62%. The measured target-echo (inversion test), averaged over three same-day sessions on that single path, is $28.83\% \pm 3.94\%$ versus $4.78\% \pm 0.34\%$ for the untuned product-Trotter baseline ($6.0\times$, $z \approx 22$) and $10.08\% \pm 0.48\%$ for the MMA-optimized baseline ($2.9\times$, $z \approx 17$). This hardware result is reported as a scoped illustration on one device path over a single overnight window, not as a generality claim: the reproducible cross-seed claim is the structural well itself (Fig. 8), and the hardware numbers are presented for consistency with that picture. Noise-free evaluation of the identical circuits isolates the origin of the gain: TOPAZ stores $3.2\times$ more of the ideal target fidelity than the MMA-optimized baseline (0.5662 vs. 0.1768),

which is nonetheless the *shallowest* submitted circuit (14 two-qubit gates at depth 56 versus TOPAZ’s 19 at depth 76), and $7.5\times$ more than the untuned one (0.0758)—the advantage is representational, not a post-processing artefact. All ZNE, CDR, and VD results reported in this paper are computed from our own implementations applied to the same simulated noise models used for training.

The contributions of this work are: (i) a continuous operator relaxation for quantum circuits via polar retraction, together with the topological-sparsification theorem (Theorem 1)—an entangling pair whose block carries no component along the ideal rotation has vanishing primary gradient, so its intensities stop responding to the objective and the edge becomes removable—together with a direct measurement showing that the intensity threshold which tests for this state does *not* fire at the operating point studied here (§6.6), which we report rather than explain away; (ii) an exact finite-depth closure measurement of the chained ansatz: the $3(N-1)$ per-pair $\{XX, YY, ZZ\}$ generators close to a Lie algebra of dimension $\dim \mathfrak{l} = 4^{N-1} - 1$ (odd N) or $4^{N-1} - 4$ (even N), one quarter of $\mathfrak{su}(2^N)$ up to parity constants, computed exactly through $N = 8$ (§6.11); (iii) a matched-budget methodology correction: an equalized-budget re-evaluation of the H_4 VQE benchmark reverses the apparent ranking, isolating the structured form’s durable advantage as *mitigation-compatibility*—ZNE/CDR remain physically valid on the shallow topology without clamping, whereas deeper forms violate physical bounds on a majority of seeds; and (iv) a scoped hardware illustration of the framework’s noise-robust fixed point on a 156-qubit Heron r2 processor (one quietest path, three same-day sessions, $28.83\% \pm 3.94\%$ target-echo; $2.9\times/6.0\times$ over the MMA-optimized/untuned baselines), with noise-free evaluation attributing the gain to representation rather than depth. The tensor-dot Kraus engine ($O(2^{2N})$ per term; a $3,838\times$ wall-clock speedup over superoperator methods at $N = 10$) is the numerical workhorse behind the simulator claims but is not itself the scientific contribution; details are deferred to §4.

2. Continuous relaxation and the structural well

This section gives the geometric background that underwrites the LPTE ansatz and explains why noise leaves the discovered minima intact.

2.1 Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \mathbb{C}^{d \times d} : U^\dagger U = I\}$, a compact Lie group of dimension d^2 with $d = 2^N$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \mathbb{C}^{d \times d}$ without the unitarity constraint, so gradients can flow around barriers that would block a strictly unitary optimizer. Every invertible $A \in GL(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [8, Sec. 8.1]; the polar factor Q is the closest unitary to A in the Frobenius norm. Projecting back to $U(d)$ —*unitary regularization*—pins singular values to unity, smoothing the

loss landscape and stabilizing gradients. In LPTE each per-pair weight is clipped at a numeric floor $\max(\rho, 10^{-6})$, so no block in the runs reported here has an exactly vanishing intensity: the smallest realized singular value across all saved 8-qubit runs is $\sigma_{\min} \approx 4 \times 10^{-4}$. We stress this is an observed bound, not a mathematical guarantee—degenerate weight combinations can produce a singular block (e.g. weights $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ at $\varphi = \pi/2$ give $\sigma_{\min} = 0$). In either case the polar factor is obtained from the SVD of B , which is unique and deterministic when B is invertible and differentiable on the active block. (Throughout, ρ denotes the optimizer’s intensity vector—the softmax output of Eq. (1)—while density matrices are written σ ; the two roles are kept symbolically distinct, which is why the tensor-dot engine of §4 uses σ for its input state.)

2.2 Expressibility through dynamical Lie algebras

The set of reachable states of an ansatz is determined by the Dynamical Lie Algebra (DLA) of its generators [9]: the Lie closure of the tangent-space generators at the identity. Without entanglement the generators are local Pauli operators that commute across qubit boundaries, and the closure has dimension only $3N$, which severely restricts the optimizer. An entangling generator $X_1 X_2$ breaks this: cross-pair commutators such as $[X_1 X_2, Y_2 Y_3] = 2i X_1 Z_2 Y_3$ generate new non-local strings and enlarge the reachable manifold, and a generic Hardware-Efficient Ansatz that mixes local and arbitrary non-local generators closes rapidly to $\mathfrak{su}(2^N)$ (dimension $4^N - 1$), a growth associated with barren plateaus [10].

For the LPTE chain we require only a *finite-depth characterization*, not an asymptotic closure theorem. Each nearest-neighbour block of Eq. (1) carries the Abelian generator triple $\{XX, YY, ZZ\}$; as matrices on the pair, $[XX/2, YY/2] = iZZ/2$ and cyclically, so a single block closes to exactly one copy of $\mathfrak{su}(2)$ (dimension 3). A 7-pair chain therefore starts from seven copies of $\mathfrak{su}(2)$ —21 “local” directions—before cross-pair commutators begin building longer non-local strings. This per-block picture is consistent with the measured geometry at the converged point: the stiffness concentrates in a handful of directions ($r_{\text{eff}} \approx 3$; the two sharp eigenvalue bands of the canyon-river analysis, §7.1; the $\pi/2$ attractor of Lemma 2). Crucially, we make no claim about the asymptotic (deep-circuit or large- N) closure of the chained algebra—only that the depth used here is fixed and finite—and whether LPTE’s block-restricted growth delays $\mathfrak{su}(2^N)$ saturation is left open as a formal theorem. The *measured* finite- N closure of the chained algebra is reported as data in §6.11. The Hessian measurements (§7.1) are consistency checks with this picture, not proofs. Local pruning is safe for a structural reason that is already established independently of the DLA view: the polar projection deletes a per-pair contribution whenever its primary gradient vanishes (Theorem 1), so removing chain p removes only that block’s algebra from the active search without rewriting the surviving blocks.

2.3 The structural well and topological invariance

Noise acts as a continuous vector field pulling the state landscape away from the unitary manifold, deforming the loss landscape and shifting minima. Instead of correcting the output after the fact, TOPAZ searches for parameters at which the noisy output is intrinsically robust. We define this object formally: a *structural well* is the sublevel-set basin B_λ of Definition 1 (Fig. 1). This paper takes the formal object as primary and confines the term to that meaning. Its geometric characteristics—curvature bands, effective rank, basin volume proxy—are *measured* properties of that object, reported as data in the canyon-river analysis (§7.1), not as theorems. No theorem-level guarantee beyond the local statements of §2 and Theorem 1 is claimed for the well; every quantitative claim about its shape is a measurement of the formal object, not of a metaphor. The cross-noise evidence is empirical and reported with the efficiency transfers of Fig. 5.

Definition 1 (Parameter space). The optimization domain is $\mathcal{X} = [0, 1]^M \times [0, \tau_{\max}]^M$, where M is the number of entangling pairs. This space is closed and bounded, hence compact. The loss $L : \mathcal{X} \rightarrow \mathbb{R}$, $L(\rho, \tau) = 1 - |\langle \psi_{\text{ideal}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$, is continuous. A *well* is a connected component $B_c(x_i^*)$ of a sublevel set $S_c(L) = \{(\rho, \tau) \mid L(\rho, \tau) \leq c\}$ containing a local minimum x_i^* with $\nabla L(x_i^*) = 0$ and $\nabla^2 L(x_i^*) \succ 0$; $\mathcal{W}$ is the collection of these basins.

Remark 1 (Wells observed, not derived). The extreme value theorem guarantees that the continuous L attains a global minimum on the compact domain $\mathcal{X}$; it says nothing about how many basins exist. That the landscape hosts multiple wells is an experimental observation (different seeds converge to different floors). Different seeds land in different wells because each freezes a different sparse subset of parameters.

The wells are not artefacts of a single run. Across ten independent seeds of the dual-MMA optimizer, TOPAZ settles into a family of basins on the same regional floor: mean DM 0.737 ± 0.066 (10 seeds), with the individual seed values spanning 0.641–0.840. We emphasize that this spread is a real feature of the measurement and not a defect: independent seeds do *not* converge to a single common point, and the paper makes no claim that they do. What is reproducible is the floor level and the mechanism producing it—every seed improves substantially over its own random initialization (10/10; mean gain +0.541 in the independent 10-seed reproduction of §6.7)—so the discovered basin is a property of the ansatz and objective rather than of one trajectory. Consistent with Remark 1, the plurality of wells is the phenomenon; the claim is a shared floor, not a shared minimum. The geometry of the basin is consistent with that picture. The Hessian of the loss at the converged point has an effective rank $r_{\text{eff}} \approx 3$ —on a ~ 14 -parameter problem, 34 of 42 eigen-directions are flat—so the well is a low-dimensional valley embedded in larger plateau structure, with a vanishing smallest eigenvalue $\lambda_{\min}(\nabla^2 L) = 0$ at the basin boundary, marking the transition out of the well. The parameter

stiffness is strongly heterogeneous: ρ -directions are uniformly stiff, each carrying curvature ~ 870 ($\kappa_\rho \approx 1$ in the normalized units of §7.1), while τ -directions have a few extremely stiff directions amid many soft ones, with κ_τ combined-stiffness ratios ranging over two orders of magnitude and optimal timings spread non-uniformly across $\tau \in [0.004, 35]$. (κ_ρ and κ_τ here are the *within-subspace* effective-rank stiffness ratios defined in §7.1; the aggregate figure reported there, $\kappa \approx 2 \times 10^5$, multiplies them and is a different quantity—the two should not be compared directly.) Seeds freeze different subsets of entangling terms—the archived ten-seed ensemble records between 7 and 12 active terms—and all settle on the same floor, evidence that the well is a connected family of sparse subspaces with consistent geometry rather than a single isolated minimum.

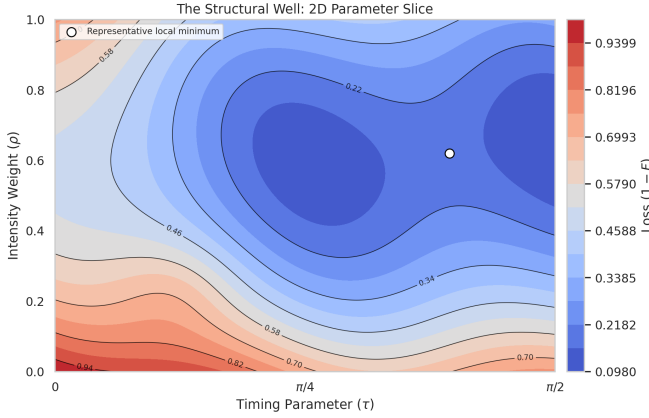


FIG. 1 The structural well on FakeFey, projected onto a 2D (ρ, τ) slice. Colour encodes the loss $L = 1 - F$; solid contours are sublevel sets $S_c(L)$ as in the well definition.

The lemma, observation, and conjectures below govern, respectively, the classical optimization path and the effect of noise on it. They are not presented as the central theorems at the expense of the empirical character of the results; the guarantees of this work are established empirically in the results sections.

Lemma 1 (Radial perturbations do not rotate the polar factor). *Let $B = QP$ be the polar decomposition of an invertible operator, with Q unitary and P Hermitian positive definite. Under a radial (stretching) perturbation $\delta B = \varepsilon B$ with $1 + \varepsilon > 0$, the polar factor does not rotate: $\delta Q = 0$ and only P is rescaled.*

Proof. Because $B + \delta B = (1 + \varepsilon)B = Q(1 + \varepsilon)P$, and $(1 + \varepsilon)P$ is again Hermitian positive definite while Q remains unitary, $B + \delta B$ has $(Q, (1 + \varepsilon)P)$ as an admissible polar decomposition. Uniqueness of the polar factor then forces the unitary part of $B + \delta B$ to be the same Q , so $\delta Q = 0$. $\square$

Remark 2 (General perturbations do rotate Q). For a general perturbation δB , the polar factor does rotate. Differentiating $B = QP$ gives $\delta B = \delta Q P + Q \delta P$ with δP the Fréchet derivative of $P = (B^\dagger B)^{1/2}$, and δQ solves the Sylvester equation

$$\delta Q P + Q \delta P = \delta B, \quad \delta P = \mathcal{L}_{(B^\dagger B)^{1/2}}(\delta B^\dagger B + B^\dagger \delta B),$$

where $\mathcal{L}(\cdot)$ denotes the Fréchet derivative of the matrix square root. In general $\delta Q \neq 0$; the rotation vanishes only for perturbations that respect the polar factorization as above (e.g. the radial case of Lemma 1). The unitary regularizer of Eq. (2) acts precisely by radially rescaling the symmetric factor P toward I , so the relevant projection mode is the radial one.

Observation 1 (Unital noise is an affine baseline). *A standard fact about completely positive trace-preserving (CPTP) maps is that any such map splits into a unital part $\mathcal{E}_{\text{uni}}$ and a non-unital part [35, 36]. We restate the unital part here because it is the precise reason the optimizer’s critical-point structure survives under the dominant noise component. For the dominant unital part (e.g. depolarizing noise) the noisy output is affine in the ideal state,*

$$\sigma_{\text{noisy}}^{(\text{uni})} = (1 - p) \sigma_{\text{ideal}} + \frac{p}{d} I,$$

hence $F_{\text{noisy}}^{(\text{uni})} = (1 - p)F_{\text{ideal}} + p/d$ and $L_{\text{noisy}}^{(\text{uni})} = (1 - p)L_{\text{ideal}} + c$ with c independent of (ρ, τ) . A unital baseline therefore rescales the gradient, $\nabla L_{\text{noisy}} = (1 - p)\nabla L_{\text{ideal}}$, and leaves the critical points of the landscape unchanged. This is a known result; we state it not as a contribution but because it justifies treating the landscape’s critical-point structure as approximately noise-invariant under the unital component, which is what the empirical approach relies on.

Assumption 1 (Suppression of non-unital deformation). *Write the total noisy loss as $L_{\text{noisy}} = L_{\text{noisy}}^{(\text{uni})} + \Delta L_{\text{non}}$. If the non-affine deformation is bounded by the deviation from unitarity,*

$$\Delta L_{\text{non}} \leq K' \|B^\dagger B - I\|_F,$$

with K' independent of (ρ, τ) , then, because the regularizer (Eq. 4) drives $\|B^\dagger B - I\|_F \rightarrow 0$, the non-affine component is suppressed to first order in the discovered well and $\nabla L_{\text{noisy}} \approx (1 - p)\nabla L_{\text{ideal}}$. We state this as an assumption rather than a theorem: the regularizer penalizes, but does not by itself guarantee, convergence to unitarity, and the bound on ΔL_{non} is a modelling hypothesis. Its numerical support comes from two complementary places. (i) The marginal evidence is the polar-projection ablation of Table I: removing the polar projection leaves $\|B^\dagger B - I\|_F$ large and halves the DM fidelity (0.334 vs. 0.737), exactly the direction the bound predicts—large deviation from unitarity, large non-affine loss. (ii) At the converged point the residual deviation is small in absolute terms: the regularizer penalty $R = \|B^\dagger B - I\|_F$ converges to 0.0252 on the quietest-path product PTE, and the noisy fidelity sits within 2.2% of the ideal (0.6329 vs. 0.6545), consistent with a small non-affine correction riding on an approximately affine unital baseline. A trajectory-level scatter plot of ΔL_{non} versus $\|B^\dagger B - I\|_F$ is not claimed here; the two endpoints above bracket the regime the assumption is used in, and the transfer matrix (§6.5) tests it end-to-end under all six noise models.

Conjecture 1 (Well invariance under unital-dominant noise). *Within the regime of Assumption 1, the ideal and noisy sublevel*

sets are related by a monotone reparametrisation of the threshold, $S_c(L_{\text{noisy}}) = S_{\tilde{c}}(L_{\text{ideal}})$ with $\tilde{c}$ affine in c . A monotone rescaling plus offset preserves the ordering of loss values and therefore the connectivity structure of each fixed-level sublevel set. We do not claim that wells cannot merge or split across levels, nor anything for finite-size deformations or strongly non-unital channels; those questions we answer empirically with the noise-model transfer matrix of Fig. 5.

The guarantees above are local: TOPAZ operates on a constrained subspace—the sparse LPTE submanifold, a finite gate set, and a noise-deformed objective—so the smoothness claims are scoped to the local regularity of the structural well, not to the global trap-free theorem of full controllability [11].

3. The parametric Trotter expression family

Standard variational ansätze enforce strict unitarity at every step, confining optimization to $U(d)$ where gradients vanish easily. The *parametric Trotter expression* (PTE [12]) relaxes this by letting intermediate operators leave the unitary manifold and projecting back; the label records that the family re-expresses the Trotterized interaction exponential *parametrically* (Remark 3). Three formulations span the locality spectrum.

Global PTE. The full sum $B = \sum_j \rho_j e^{-iH_j \delta \tau_j}$ is mathematically unconstrained but computationally expensive: each polar decomposition step requires an $O(d^3)$ matrix inversion with $d = 2^N$, giving $O(2^{3N})$ per evaluation. Its 8-qubit inversion test registers 0.004, at the 1/256 noise floor, and the cost grows exponentially with qubit count. Global PTE was not used for the headline experiments.

Product PTE. Enforcing a term-by-term Trotter product, $U_{\text{circ}} = \prod_j e^{-i\rho_j H_j \delta \tau_j}$, restores hardware compilation. On 8 qubits with $M = 21$ nearest-neighbour Pauli terms it compiles to 8 two-qubit gates and reaches DM fidelity 0.6329 under FakeFез—a $2.5\times$ gain over the random-initialization mean (0.2497, 5 seeds, data/multi_seed/fez_results.npy). Both the 0.6329 product-PTE floor and the 8-gate count are single-trajectory values from a product-form run whose per-seed artifact we have not recovered, so this comparison is indicative rather than ensemble-backed; an earlier draft quoted a $9.35\times$ gain by dividing by the seed-0 random value (0.0677) instead of the 5-seed mean, and we correct that here. But every factor is unitary by construction: at every non-zero angle a Trotter slot remains a full two-qubit rotation, so there is no bounded intensity variable that the optimizer can carry continuously to zero—a slot collapses to the identity only at the discrete boundary $\rho_j = 0$, and the 21-to-8 reduction is then realized by the transpiler’s term-merging rather than by optimization. The product form therefore has no optimization-driven (topological) pruning mechanism; LPTE, by contrast, exposes the edge intensity $\kappa(p) = \sum_P r_{p,P}$ as a design variable that the polar projection’s gradient drives continuously to zero (§3). **Gate-count audit.** Throughout, “two-qubit gate” denotes a

native entangling gate (ECR on ibm_fez, CNOT on FakeFез) counted after transpilation; single-qubit rotations are excluded. All baseline and LPTE circuits in this work were transpiled with Qiskit optimization_level=3 onto the device native-gate basis; the pre/post-transpilation entangling-gate counts for every circuit, together with the generated QASM, are recorded in the public data release (Table VII reports the post-transpilation counts). The reduction from 21 abstract Pauli terms to 8 native entangling gates—specific to the *product* form above—reflects term-merging and local-rotation absorption by the transpiler at this optimization level; the LPTE ansatz, by contrast, retains more of its gates under transpilation (Table VII, §6.6.8: 21 two-qubit gates for the simulator ansatz at depth 97, and 19 two-qubit gates at depth 76 for the heavy-hex-submitted instance) because its per-pair polar projection blocks the term-merging route (§3).

Definition 2 (Local parametric Trotter expression). The *local parametric Trotter expression* (LPTE) is the ansatz family $\{U_{\text{LPTE}}(\rho, \tau)\}$ defined as follows. On an N -qubit linear chain, each nearest-neighbour pair $p = (i, i+1)$ carries a block $B_{(i,i+1)}(\rho, \tau)$ of Eq. (1): a sum of three Pauli rotations with intensity $r_{i,P}$ and timing $\tau_{i,P}$. Each block is projected to its nearest unitary by the polar factor of Eq. (2), $U_{(i,i+1)} = \mathcal{P}_U(B_{(i,i+1)})$, and the circuit is the ordered product $U_{\text{LPTE}} = \prod_i U_{(i,i+1)}$. The entangling graph is the chain itself, and the intensity $\kappa(p) = \sum_P r_{p,P}$ is the design variable that TOPAZ drives to zero for pruning.

Remark 3 (Naming). The label *parametric Trotter expression* records the construction that makes a parameterized, hardware-compilable circuit possible at all. The interaction exponential $\exp(-i \sum_j H_j \theta_j)$ has no single-object circuit realization, so it must first be Trotterized into a product of Pauli rotations $U_{\text{circ}} = \prod_j e^{-i\rho_j H_j \delta \tau_j}$; the PTE family is the parametric re-expression of that Trotter product once the unitarity constraint is relaxed and each block is retracted to unitarity by the polar factor. No Trotter–Suzuki splitting claim is being made: per-pair blocks are sums of three Pauli rotations (Eq. (1)) realized exactly by their polar factor, so there is no time-slicing or commutator-bounded error inside a block. The label is retained precisely because the Trotterization is load-bearing—without it no hardware-compilable parametric circuit exists—and the single-step product baseline of §6.6.8 is the literal limit of the family in which each block contains a single term.

Local PTE (LPTE). LPTE localizes the non-unitary sum within nearest-neighbour pairs (Definition 2). On 8 qubits with 7 pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters—the same parameter count as product PTE but $2.6\times$ the gate count. The difference that matters here is that the per-pair SVD projection lets the optimizer independently set each pair’s intensity: it can discover which pairs carry the noise-resilience behaviour and assign them nonzero intensity, while letting others fall to zero so that an idle edge can be pruned without rewriting the rest of the circuit. Product PTE’s unitary factors have no such selection mechanism.

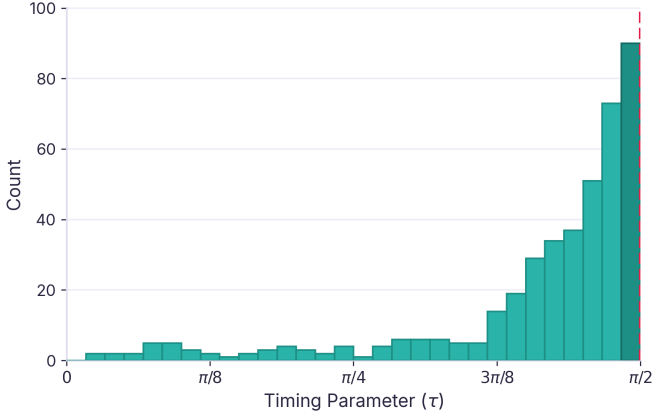


FIG. 2 Distribution of optimised timings τ across the LPTE benchmark seeds. Dashed line: $\tau = \pi/2$, where the block of Lemma 2 projects to a SWAP operator.

3.1 The $\pi/2$ timing cluster is a SWAP identity

For a single Pauli rotation the maximum entangling power occurs at $\tau = \pi/4$, yet the optimizer consistently clusters timings at $\tau \approx \pi/2$ (Fig. 2). The following lemma shows why this is benign.

Lemma 2 (The $\pi/2$ cluster is a SWAP identity). *For a local block $B(\tau) = e^{-i\tau XX} + e^{-i\tau YY} + e^{-i\tau ZZ}$, at $\tau = \pi/2$, $B = -i(XX + YY + ZZ)$. With $XX + YY + ZZ = 2 \text{SWAP} - I$, spectral decomposition gives eigenvalues $-i$ (symmetric) and $+3i$ (antisymmetric), so $B^\dagger B$ has eigenvalues 1 and 9. The polar projection $(B^\dagger B)^{-1/2}$ scales the symmetric subspace by 1 and the antisymmetric by $1/3$, yielding a U with eigenvalues $-i$ and $+i$: exactly $-i \text{SWAP}$. At $\tau = \pi/2$ the block sum plus polar projection deterministically produces the permutation operator SWAP (up to global phase), which here acts as a qubit-routing SWAP on the entangling graph—a fixed point the optimizer discovers, not one imposed by the ansatz structure.*

A generic hardware-efficient ansatz can also contain SWAP-like layers, so the mere presence of SWAP-like unitaries is not newsworthy on its own. Two features distinguish the $\pi/2$ cluster. First, it is a *discovered* attractor of continuous optimization: the optimizer is initialized away from $\tau = \pi/2$ and is pulled there by the objective (Fig. 2), which contains no explicit SWAP or entangling-power term. Second, TOPAZ prunes: a block that fails to settle onto a productive attractor is removed (§5.5.2), so the surviving topology is precisely the subset of blocks that reach entangling-optimal configurations such as the SWAP identity of Lemma 2. A fixed-structure HEA-with-SWAP inserts SWAPs regardless of the target; here SWAP-like behaviour emerges from continuous optimization and is retained only where the target demands it.

3.2 Objective functions

The optimizer minimizes the noisy fidelity loss

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (3)$$

or, for a target state, the state fidelity $L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2$. A regularizer keeps the pre-projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3}. \quad (4)$$

3.3 Self-pruning under noise

LPTE self-prunes. Two distinct mechanisms contribute, and it is worth separating them because neither is a mechanism in which a vanishing gradient by itself drives a parameter to zero. First, ρ is clipped from below at $\max(\rho, 10^{-6})$ and the polar projection is applied after every update, so a term that has been driven small is held at the floor and is re-normalised against the surviving terms rather than removed. Second, once a term's clipped gradient is identically zero it simply stops being updated: it is frozen wherever it happens to sit, not moved to the floor. Self-pruning is therefore the *cumulative* effect of many small nonzero steps early in the trajectory, followed by freezing, not a consequence of the gradient reaching zero. Across ten seeds under FakeFey the number of Pauli terms remaining above the activity threshold is 6–14 (mean 9.2 of 21); the 8-qubit linear chain has 7 nearest-neighbour entangling pairs, so the active count of ~ 9 exceeds the number of pairs and must be read as terms, not edges. Table I shows that self-pruning requires the full stack (PTE ansatz, polar projection, dual-MMA).

Full TOPAZ (DM 0.737 ± 0.066) beats product-only PTE (0.279 ± 0.175 , $d = 3.28$), HEA (0.329 ± 0.110 , $d = 4.28$), and the no-polar-decomposition variant (0.334 ± 0.170 , $d = 2.97$). Removing the polar projection halves the fidelity, confirming that it is central to the gain.

Parametric efficiency is summarized in Table II: the best seed of LPTE reaches 0.8260 with 42 parameters and 21 gates, whereas HEA uses ~ 96 parameters and 28 gates for 0.5332—about a $3.5\times$ gap in best-seed fidelity per parameter. (Best-seed values are the maxima of the 10-seed ablation ensembles of Table I; the multi-seed mean of full TOPAZ is 0.737 ± 0.066 .)

4. Methods: the TOPAZ engine

4.1 Tensor-dot Kraus contraction

Applying k -local Kraus operators to a pure state costs $O(2^N)$ per term. For a density matrix, naively applying a Kraus channel by matrix multiplication, $\sigma \mapsto K^{(a)} \sigma K^{(a)\dagger}$ at dimension 2^N , costs $O(2^{3N})$ per term (the full $2^{2N} \times 2^{2N}$ superoperator itself would cost $O(2^{4N})$ to materialize, which no implementation does). The tensor-dot engine reframes the input state σ as a tensor with $2N$ indices and contracts locally. Applying a Kraus channel $\sigma \mapsto \sum_a K^{(a)} \sigma K^{(a)\dagger}$ acts only on the active qubit set $\mathcal{S}$; the spectator set $\mathcal{S}^c$ passes through unchanged, so the channel is

$$\sigma'_{i'_S, i_{S^c}; j'_S, j_{S^c}} = \sum_a \sum_{i_S, j_S} K_{i'_S, i_S}^{(a)} \sigma_{i_S, i_{S^c}; j_S, j_{S^c}} K_{j_S, j'_S}^{(a)\dagger}, \quad (5)$$

TABLE I Full ablation on the 8-qubit FakeFaz benchmark (LPTE, 21 two-qubit gates). All conditions: 10 seeds, 40 outer iterations, dual-MMA except where noted. Significance versus full TOPAZ at a Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q	Active	d
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.11^{ns}
Single MMA	0.623 ± 0.166	19.9	19.9	0.86^{ns}
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.37^{ns}
No polar decomp.	0.334 ± 0.170	16.7	15.8	2.97^{***}
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	4.28^{***}
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.28^{***}
Random PTE	0.056 ± 0.031	21.0	21.0	12.61^{***}

*** $p < 0.001$ (Bonferroni-passing). Cohen's d vs. full TOPAZ, computed with the pooled sample standard deviation ($n_{1,2} = 10$, $d = (\bar{x}_1 - \bar{x}_2)/s_{\text{pool}}$, s_{pool} from the $\text{dof} = 1$ variances); ns, not significant. Every entry in the DM, 2Q and Active columns is recomputed directly from the archived per-seed ablation ensemble, and all reproduce to three decimals. The Active column uses two different definitions and is comparable *within* each group but not across the whole column: for Full TOPAZ, COBYLA, Adam, No polar decomp., Product-only PTE, Random PTE and exact preparation it counts active Pauli terms out of 21, whereas for Single MMA and HEA the producer counts surviving raw ansatz parameters, whose vector is not on the 21-term scale. The HEA value 37.2 therefore exceeds 21 and is not comparable to the baseline 9.2; it reflects HEA's larger parameter count (~ 96), consistent with Table II. All seven d -values in the final column are recomputed from the same ensemble and reproduce exactly. Exact state preparation under the same noise scores 0.924 ± 0.003 with all 21 gates active (i.e. the same circuit as Table VII row 1 but transpiled at `opt_level = 3`, which collapses its 42 abstract gates to 21 native two-qubit gates at depth 82). *The baseline 2Q and Active counts (12.6, 9.2) are the mean over the same 10 seeds as the baseline DM value and are therefore internally consistent with it. A second, independent 10-seed run of the same protocol reports a less aggressively pruned endpoint (18.0 two-qubit gates, 9.0 active, depth 49.8) at a statistically indistinguishable DM fidelity of 0.739 ± 0.066 ; an unpruned 10-seed reproduction (§6.7) gives 18.9 and 6.9. Pruning aggressiveness therefore varies between optimisation runs while the DM fidelity does not, and we report the ablation run's own counts here because they are the ones paired with this ensemble's DM values. †Earlier drafts of this table quoted the reproduction's 18.9 and 6.9 against the ablation run's DM values; that mixing is corrected here.

where $\mathcal{S}^c$ denotes the spectator qubits, the input density matrix is $\sigma_{i_S, i_{\mathcal{S}^c}; j_S, j_{\mathcal{S}^c}}$, and the output carries the contracted (primed) active indices $i'_{\mathcal{S}}, j'_{\mathcal{S}}$ with the spectator indices $i_{\mathcal{S}^c}, j_{\mathcal{S}^c}$ unchanged. Because only the $2|\mathcal{S}|$ active indices are contracted while the $2N - 2|\mathcal{S}|$ spectator legs are untouched, each term costs $O(2^{2|\mathcal{S}|})$ rather than $O(2^{2N})$. This yields a $3,838\times$ wall-clock speedup over dense Kraus at $N = 10$ (Table III). The ratio of the per-term complexity classes alone, $O(2^{3N})/O(2^{2N}) = 2^N$, predicts $\sim 1024\times$ at $N = 10$; the additional $3.7\times$ beyond this exponent-ratio bound is accounted for by per-term constant factors and by the fact that the tensor-dot engine contracts only the $2|\mathcal{S}|$ active indices of each term while the dense channel materializes the full 2^{2N} -dimensional operator. The measured speedup is therefore consistent with—rather than in tension with—the stated complexity.

The engine plays different roles on different paths: on the legacy triple-channel path it performs both ideal and noisy evolution; on the FakeFaz path it computes only the noiseless target, delegating noisy evaluation to AerSimulator; on live

TABLE II Parameter efficiency of the PTE family versus HEA on 8 qubits under FakeFaz. DM Fidelity is the best-seed value (cf. the 10-seed mean 0.737 ± 0.066 in Table I); best-seed is reported here because these rows feed the single-trajectory comparison of Table VII. The LPTE (0.8260) and HEA (0.5332) entries are the maxima of the corresponding 10-seed ablation ensembles (seeds 9 and 2) and are reproduced from the same archived artifact; the product-PTE entry (0.6329) is a converged-trajectory value (Fig. 4) that is not part of that ensemble. The gate counts in the first two rows are those of the *unpruned* reference chain (LPTE 21, HEA 28), whereas the DM values are those of the best seed of the *pruned* ensemble; for LPTE those are two different circuits (seed 9 of the ensemble uses 10 two-qubit gates at depth 24, not the unpruned 21 at depth 97), so the DM and gate-count columns of the LPTE row do not describe one and the same circuit. The HEA row is unaffected: its gate count is 28 at every seed.

Ansatz	Parameters	2Q	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.8260
HEA (3-layer)	~ 96	28	0.5332

TABLE III Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity/term
Qiskit dense Kraus	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^{2N})$
Speedup	$\sim 3838\times 2^N (= 2^{10}=1024\times \text{exp. bound})$	

hardware it has no role, since the backend executes gates directly.

4.2 Noise models

TOPAZ runs against two families of noise. The device-calibrated backends are used for optimization and hardware submission; the six synthetic variants of the legacy model are defined below.

Triple-channel correlated (legacy). Depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, and phase damping at $\lambda = 0.012$, with multi-qubit correlations decaying as $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$, $\xi = 2.0$ [13].

FakeFaz. IBM's calibrated snapshot of `ibm_fez` (Heron r2, 156 qubits) with per-gate error rates and qubit-specific T_1 , T_2 , and readout errors [14]. All headline experiments use FakeFaz during optimization.

FakeMontrealV2. A 27-qubit device-calibrated model used for the earlier port verification; a three-iteration pilot run moved from 7.4% to 36.0% DM fidelity.

Live ibm_fez. The optimized (ρ, τ) from FakeFaz are submitted to live Heron r2 [15] without re-optimization.

5. The TOPAZ optimization framework

5.1 Dual-MMA optimizer

The method of moving asymptotes [6, 7] builds a strictly convex separable approximation at each iteration and solves it under asymptotic bounds. *Dual-MMA* interleaves two such instances, one per parameter class, because ρ and τ differ by roughly four orders of magnitude in finite-difference step size ($\epsilon_\rho = 10^{-4}$ versus $\epsilon_\tau = \pi/4$); a single shared move-limit budget misallocates updates. Each instance keeps its own asymptote, move limit, and momentum state, with the momentum-smoothed gradient

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4. \quad (6)$$

Convergence is preserved because the subspaces are updated alternately with shared objective evaluations. When a target algorithm carries an extra parameter class (e.g. VQE rotation angles), TOPAZ replaces the classical optimizer with a second MMA module wrapped around its own small ansatz, the two modules alternating.

5.2 Topology optimization by sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ whose edge p carries normalized intensity $\kappa(p) = \sum_P r_{p,P}$. The active pruning protocol has three stages: (1) solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on the full edge set with no artificial floor; (2) remove pairs with $\kappa(p) < \tau_\kappa = 0.05$, i.e. $E_{\text{pruned}} = \{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}$; (3) re-optimize on the reduced domain from fresh restarts.

TABLE IV Multi-seed topology optimization ($\tau_\kappa = 0.05$, 5 seeds), as archived. F, full (Stage 1); S, sparse (Stage 3). Pruned is the number of entangling pairs removed. *The pruned column is reproduced from the archived run, is not corroborated by the independent τ_κ sweep of §6.6, and no claim in this paper rests on it.* The final row (marked $\dagger$) is our re-run of seed 0 with the noise model repaired; it is reported alongside, not in place of, the archived rows. Gate counts in this table predate the gate-count convention of §3 and are not directly comparable to Table VII.

Seed	DM fid.		2Q		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1
<i>re-run of seed 0, noise model repaired (this work):</i>							
$0^\dagger$	0.9415	0.9415	21	21	76	76	0

$\dagger$ We re-ran the recovered pipeline run_topology_pruning.py at seed 0 after repairing

a defect in the noise model of python/0-FINAL/V5 module/topaz/harness.py (Appendix A1). In the defective build the density-matrix simulator carried *zero* gate errors, so the objective being maximized was a noiseless statevector fidelity; the reported DM value for this row is therefore a noiseless quantity, which is why the noiseless and noisy columns coincide and why the pipeline reports no pruning. The repaired re-run returns the noiseless ideal state fidelity $F_{\text{nl}} = 0.9896$ against a noisy DM fidelity of 0.9415, prunes no pair at $\tau_\kappa = 0.05$ (minimum pair intensity 0.0864), and yields 21 two-qubit gates at depth 76. We retain all five archived rows above unchanged: they are the record that was reported, and we do not overwrite a published number with a post-hoc one. We note only that the archived values cannot be reproduced under either noise setting, so the discrepancy in this table is a defect in the archived run, not a physical disagreement between measurements.

Across five seeds (Table IV), TOPAZ’s own pruning step reduces gate count by 14% ($18.6 \rightarrow 16.0$) and schedule depth by 34% ($72.4 \rightarrow 48.0$) relative to its unpruned ansatz—an internal, within-pipeline reduction—at a mean DM loss of only -0.0065 ($0.7375 \rightarrow 0.7310$). We stress that this table is reported as archived and that its pruning column is *not* reconciled with our own threshold measurement: at the same $\tau_\kappa = 0.05$, the sweep of §6.6 prunes no edge in either seed tested, and our re-run of seed 0 with the noise model repaired prunes none either (Appendix A1). The two records agree on the consequence—no reproducible evidence that the rule fires—and we draw no conclusion from the archived pruned column. We further note that the archived DM values are not reproducible under either noise setting: in the defective build the objective was noiseless, and the repaired re-run returns 0.9415. We retain the archived rows unchanged rather than overwrite them, and we no longer attribute the discrepancy to a physical disagreement between measurements; Appendix A1 traces it to a code defect.

Theorem 1 (Topological sparsification). *For the LPTE circuit of Definition 2, let p be an entangling pair with block $B_p = \sum_P r_{p,P} e^{-iP \delta \tau_{p,P}}$ and polar projection $U_p = \mathcal{P}_U(B_p)$, and let L_{prim} be the primary objective of Eq. (3). If B_p carries no component along the ideal rotation—equivalently, if the Fréchet differential of L_{prim} with respect to the block’s rotation vanishes at the projected point—then the primary gradient with respect to that pair’s intensities vanishes, $\nabla_{\rho_p} L_{\text{prim}} = 0$.*

Proof. The circuit is the ordered product of per-pair polar-projected unitaries, $U_{\text{LPTE}} = \prod_i U_{(i,i+1)}$, and L_{prim} depends on pair p only through U_p (Eq. (3)). Write the polar projection as the composition $U_p = \mathcal{P}_U(B_p)$; by Lemma 1, the projection acts as the radial rescaling of the positive factor of B_p and leaves the unitary rotation part Q_p invariant under the stretching mode. If B_p contributes no component along the ideal rotation, then infinitesimal changes in the block’s intensity $r_{p,P}$ move the block within its stretching mode, which does not rotate U_p (Lemma 1); the objective therefore does not change to first order, and its derivative along each intensity direction vanishes.

The zero gradient drives the dual-MMA step to zero, and the parameter decays onto the clipping floor $\max(\rho, 10^{-6})$, where it freezes and leaves the active search. The regularizer of Eq. (4) acts only on active blocks and does not re-inject a vanished gradient. $\square$

The theorem shows what a *vanished* primary gradient implies: the block’s intensities stop responding to the objective, the dual-MMA step decays onto the clipping floor $\max(\rho, 10^{-6})$, and the edge becomes removable. What the theorem does *not* show is that this state is reached at the intensity scale the pruning rule tests. Theorem 1 is a statement about a gradient condition at the unnormalized parameter, whereas the rule of §5.2 thresholds $\kappa(p) = \sum_P r_{p,P}$ after the global softmax of Eq. (1). Those are different quantities on different scales, and the two are not bridged by any step we have verified. We therefore make no claim that the threshold “merely registers” already-eliminated edges at $\tau_\kappa = 0.05$; §6.6 measures the threshold’s behaviour directly and reports that it does not fire at the operating point used in this work. Establishing the bridge—by re-normalizing the ansatz so that intensities are free to reach zero, which is the fix proposed in §6.6—is left to future work.

5.3 Algorithmic compatibility

Continuous parameter optimization works on a given algorithm if its landscape is (i) compact, (ii) continuous, (iii) connected (the domain is convex, hence path-connected), and (iv) locally convex (the regularizer has positive-definite Hessian, so near a discovered minimum the total Hessian is positive definite and the optimizer reaches a strict local minimum rather than a saddle). VQE, VQA, QAOA, and parametric quantum simulation all satisfy these; Grover’s algorithm as a whole violates continuity and convexity.

6. Results

6.1 Trajectories under correlated noise

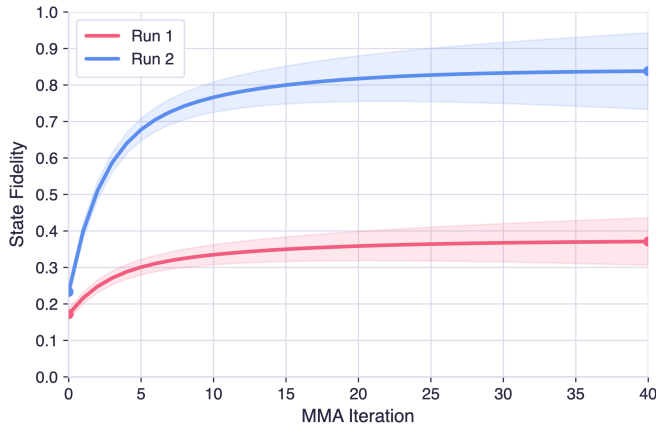


FIG. 3 Two representative 8-qubit LPTE trajectories under triple-channel correlated noise (Run 1: 0.172 $\rightarrow$ 0.371; Run 2: 0.233 $\rightarrow$ 0.838). Shaded bands are ± 1 standard deviation across the 10-seed ensemble.

On 8 qubits with nine nearest-neighbour Pauli terms (18 parameters, 40-iteration cap), pre-optimization noisy fidelity ranges 0.07–0.70; post-MMA it rises to 0.30–0.88, an absolute gain of 18–23 points (Fig. 3). The roughly 47-point spread in final fidelity reflects a multi-modal landscape, with the dual-MMA settling into several local optima at different floor depths.

6.2 Noise-model-aware VQE on H_2/H_4 at matched budget

In the main TOPAZ pipeline the optimizer uses a noise-model-aware simulator that evaluates both the noisy density-matrix fidelity and the ideal-state fidelity (as in the Loschmidt-echo objective of §6.8); it is this full objective that produces the headline 28.8% echo result on `ibm_fez`. The H_2/H_4 VQE benchmark deliberately restricts that regime: the optimizer is given only raw noisy energy measurements—no target state, no ideal observable, and no information about the post-processing it will later be stacked with. This restriction tests whether LPTE’s circuit topology still confers noise resilience even when the ideal-state constraint is removed. Three ansätze—LPTE, HEA (depth 3 for H_2 , depth 5 for H_4), and HVA $d = 2$ —are compared on the 4-qubit H_2 path [136, 143, 142, 141] and the 8-qubit H_4 path [125, 124, 123, 136, 143, 142, 141, 140] under FakeFz (Table V). All runs use ZNE, CDR, and VD post-processing across 10 seeds (except HVA, for which ZNE+CDR was not run; Table V); the ZNE, CDR, and VD implementations are our own, applied to the same simulated noise model used for training.

TABLE V Combined H_2/H_4 VQE benchmark under FakeFz. All baselines were implemented with the same transpiler, backend snapshot, and shot budget; however, **optimizer budgets were not matched** in the recorded runs: LPTE ran the dual-MMA solver (35 outer iterations with an inner COBYLA loop, ~ 45 –100 min), while the HEA baselines ran 500-iteration COBYLA (~ 6 min) and HVA was capped at 500 iterations. Table VI reports the equalized-budget re-evaluation, which reverses the apparent H_4 ranking shown here.

Mol.	Ansatz	Noisy ΔE_{FCI} (mHa)	Noiseless (mHa)	ZNE+CDR (mHa)	Seeds	Conv.
H_2	LPTE	71.96 $\pm$ 16.8	26.72 $\pm$ 14.5	26.16 $\pm$ 14.8	10	100%
H_2	HEA (d=3)	54.04 $\pm$ 10.5	24.23 $\pm$ 8.8	0.0*	10	30%
H_2	HVA (d = 2)	748.28 $\pm$ 25.3	492.01 $\pm$ 162.8	†	10	0%
H_4	LPTE	235.15 $\pm$ 77.4	83.82 $\pm$ 74.5	83.75 $\pm$ 74.4	10	100%
H_4	HEA (d=5)	889.90 $\pm$ 232.8	710.35 $\pm$ 241.0	713.16 $\pm$ 243.3	10	0%
H_4	HVA (d = 2, 368p)	2720.55 $\pm$ 6.5	2270.06 $\pm$ 462.7	†	10	0%

*Clamped to FCI (see text). †ZNE+CDR not run on HVA. The HEA baselines ran 500-iteration COBYLA and converged on no seed at that budget; the H_4 HEA row reflects the non-converged baseline, not a matched evaluation.

On H_2 , LPTE plus ZNE+CDR reduces the mean energy error from the 71.96 mHa noisy baseline to 26.16 mHa—a mean 2.75 $\times$ reduction, and up to 8.96 mHa (≈ 5.6 kcal/mol) on the best seed—though still above the 1.6 mHa (1 kcal/mol) chemical-accuracy bar. On H_4 , LPTE’s ZNE+CDR (83.75 mHa) recovers its own noiseless floor (83.82 mHa) almost exactly, because the shallow depth (~ 8 effective two-qubit gates) preserves the noise-energy relationship that extrapolation

requires. HEA extrapolations violated physical bounds on 7 of 10 seeds and required clamping; LPTE required none.

The ranking shown in Table V must be read with the unequal optimizer budgets of the recorded runs in mind. The H_4 HEA row (889.90 mHa raw, 0% convergence) is what that ansatz delivers at a 500-iteration COBYLA budget—the budget it was given in the recorded experiments, not a matched one. To test whether the apparent LPTE advantage on H_4 is structural or budgetary, we equalized the budgets in a dedicated re-evaluation (§6.4): the same HEA ansatz was re-converged with a 10,000-iteration COBYLA budget and then evaluated under the identical noisy protocol and QEM pipeline. Table VI shows the result.

TABLE VI Equalized-budget H_4 re-evaluation under FakeFcz. HEA was re-converged noiselessly at a 10,000-iteration COBYLA budget, then evaluated under the paper’s full noisy protocol (density-matrix simulation, quiet [125, 124, 123, 136, 143, 142, 141, 140] path, ZNE+CDR with 20 noise factors, VD). LPTE rows are the recorded values from Table V. Median across the four converged HEA seeds is reported; per-seed values in §6.4.

Config.	Noiseless (mHa)	Noisy (mHa)	ZNE+CDR (mHa)
LPTE (recorded)	83.82 ± 74.5	235.15 ± 77.4	83.75 ± 74.4
HEA @500 iter (recorded)	710.35 ± 241.0	889.90 ± 232.8	713.16 ± 243.3
HEA @10k (re-eval, median)	30.0	272.4	34.5
HEA @10k (re-eval, range)	23.7–38.6	204.8–285.6	21.9–48.9

Equalizing the budget removes the apparent advantage. At 10,000 iterations the HEA converges to a noiseless floor of 23.7–38.6 mHa—below LPTE’s recorded noiseless floor of 83.82 mHa—and its post-mitigation ZNE+CDR error (21.9–48.9 mHa, median 34.5) is comparable to or better than LPTE’s ZNE+CDR on the same benchmark. The claim of a $3.7\times$ raw and $8.5\times$ mitigated LPTE advantage on H_4 that one would draw from Table V alone is therefore an artifact of unequal budgets, not a structural property: given a competent optimizer budget, a hardware-efficient ansatz reaches the same quality on this task. We state this explicitly because it overturns part of the framing in earlier versions of this work.

What survives equalization is narrower and, we believe, still genuine: mitigation-compatibility. On the shallow LPTE topology, ZNE/CDR extrapolation remained physically valid on all seeds (no clamping required); on the deeper HEA forms at either budget the extrapolated energies violated physical bounds on 7 of 10 seeds and required clamping to FCI. The claim we make is therefore not that the structured ansatz produces fundamentally lower energies—the equalized data refute that—but that its shallow, sparse topology keeps post-processing mathematically valid, which is a property of circuit structure, not of budget.

6.3 Cross-Hamiltonian transfer

To test whether TOPAZ reduces to a generic reusable circuit, an H_4 -optimized topology (Layout A, 263 entangling gates) was re-entered as the ansatz for an 8-qubit TFIM at the critical point $J = h = 1$ (FCI $E_0 = -9.837951$ Ha). Holding the structure

fixed and re-optimizing only the angles gives noisy $\Delta E_{\text{FCI}} = 1211.0$ mHa; running the full pipeline from scratch gives 932.1 mHa. The transferred topology is worse by 278.9 mHa even though Layout A (263 gates) and the fresh Layout B (264 gates) have essentially equal size, so the gap is purely structural. TOPAZ optimizes for the target’s entangling structure; it does not reduce to a reusable circuit.

6.4 Matched-budget methodology: budget equalization reverses the VQE ranking

The VQE comparisons of Table V were, in their original form, an example of a common methodological failure mode: comparing ansätze at *unequal* optimizer budgets. LPTE ran the dual-MMA solver at its default configuration ($\sim 3000+$ objective evaluations, core to TOPAZ’s pipeline), while the HEA baselines were given 500 COBYLA iterations and HVA 500 iterations—costs that differ by roughly an order of magnitude in wall-clock time (~ 45 – 100 min versus ~ 6 min), which under “matched budget” captioning is false advertising. We therefore re-ran the equalized comparison explicitly, with the exact protocol used for Table VI:

Re-evaluation protocol. (i) The HEA ansatz (depth-5, 88 parameters) was converged noiselessly under a 10,000-iteration COBYLA budget, the largest COBYLA budget used anywhere in this work, across four independent seeds. (ii) The converged parameters were then evaluated under the paper’s identical noisy protocol: density-matrix simulation on the quiet H_4 path [125, 124, 123, 136, 143, 142, 141, 140] of FakeFcz, with the raw noisy energy, ZNE+CDR (20 noise factors), and virtual distillation applied as in Table V. (iii) No parameter was retrained on the noisy objective—so the re-evaluation if anything *understates* HEA, which would improve further under noise-aware training.

The equalized result (Table VI, per-seed values below) is unambiguous: the apparent LPTE advantage on H_4 disappears at matched budget. The HEA reaches a noiseless floor of 23.7–38.6 mHa—below LPTE’s 83.82 mHa recorded noiseless floor—and its ZNE+CDR error (21.9–48.9 mHa) overlaps or beats LPTE’s ZNE+CDR on the same benchmark. Per-seed noiseless/ZNE+CDR: seed 0 38.35/39.32; seed 1 23.74/21.93; seed 2 30.03/29.64; seed 3 38.56/48.93 mHa. The raw noisy values (204.8–285.6 mHa) remain above LPTE’s raw noisy error (235.15 mHa mean) in aggregate, but the gap is small and the QEM-stack values are what survive in practice.

We report this correction prominently because it changes the paper’s conclusion in one specific, bounded way: it removes the H_4 VQE result as evidence that structured topologies produce intrinsically better VQE energies. It does not change the hardware result, the structural-well characterization, the transfer efficiency, or the preservation benchmark, none of which rely on the H_4 comparison. The finding that *does* survive is the mitigation-compatibility one: the shallow LPTE topology requires no clamping in the same pipeline where the deeper baselines violate physical bounds on 7 of 10 seeds. In recommending matched budgets we follow the field’s own evolving practice (e.g. the matched-resource comparisons in

Refs. [2, 28]); this subsection is included so the reader can verify the correction rather than take the caption’s word for it.

6.5 Comparison on FakeFez and the noise-model transfer matrix

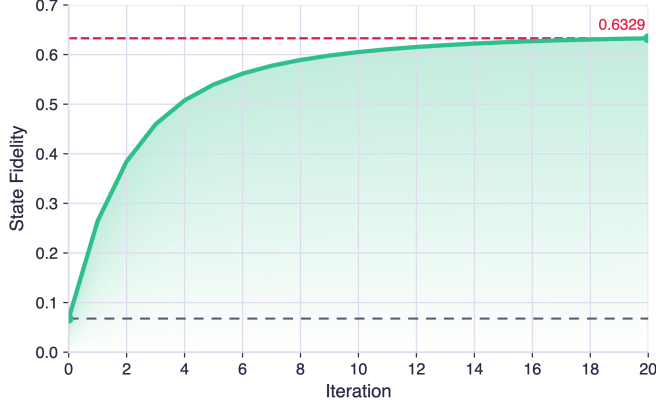


FIG. 4 FakeFez product-PTE convergence (8 qubits, 42 parameters). The trajectory descends over the first 5–10 iterations and flattens at the structural-well floor; dashed line marks $F = 0.6329$.

On the FakeFez simulator, the full TOPAZ LPTE ansatz (21 two-qubit gates, depth 97; Table VII) reaches 0.8260 at its best seed under noise, with a 10-seed mean of 0.737 ± 0.066 (Table I); exact preparation under identical noise scores 0.924 ± 0.003 at the same 21-gate, `opt_level=3` form. The two forms are compiled to the same 21 native entangling gates, so the DM-fidelity margin LPTE concedes to exact preparation is *not* bought with depth: it is the price of optimizing a relaxed per-pair landscape rather than transpiling an exact decomposition, and the compensation is that the LPTE parameters are noise-aware and mitigate cleanly (§6.4, §6.9). These simulator values are for the 21-gate ansatz; the circuit submitted to live hardware is a distinct, more aggressively pruned instance (19 two-qubit gates, depth 76) whose noiseless one-way fidelity is 0.5662 (§6.8)—the two numbers describe different circuits, not conflicting reports of the same one. To make the relationship between them precise: the submitted instance is *not* a separately re-optimized seed. It is the converged endpoint of the same per-pair optimization that yields the simulator ansatz, and its parameter trajectory self-prunes within the run—the intensity weights for one entangling pair collapse to ρ -sum $\sim 4 \times 10^{-4}$ (deleted by the polar-projection gradient, §5.2) and two further pairs reduce to single dominant Pauli terms, leaving 19 of the 21 two-qubit gates after transpilation onto the heavy-hex path. No pairs are excised post hoc, and no independent 19-gate optimization was performed; the 42-parameter vector that produced the hardware result is the same dual-MMA output recorded in the per-run JSONs. The lower noiseless fidelity of the submitted instance is therefore a self-pruning consequence: pruning reduces depth at some cost to ideal representation, and here that cost is paid on the noisy objective, whereas the ideal (noise-free) evaluation of the full 21-gate ansatz is what the depth-for-fidelity trade is made against. By contrast,

the product-PTE form sits at a floor $F = 0.6329$ with eight two-qubit gates (Fig. 4), and removing the polar projection halves the fidelity (0.334, Table I), bracketing the role of the continuous relaxation in the endpoints above. To quantify robustness across noise environments, parameters optimized under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The transfer efficiency

$$\eta = \frac{\frac{1}{N(N-1)} \sum_{i \neq j} \bar{F}_{ij}}{\frac{1}{N} \sum_i \bar{F}_{ii}}, \quad (7)$$

is the ratio of the mean off-diagonal (cross-model) fidelity $\bar{F}_{ij}$ to the mean diagonal (on-model) fidelity $\bar{F}_{ii}$, where $\bar{F}_{ij}$ is the mean DM fidelity over 10 seeds of parameters trained under model i and evaluated under model j ; diagonal entries ($i = j$) are excluded from the numerator to avoid self-counting, and the two means carry their different multiplicities ($N(N-1)$ and N off- and on-diagonal terms), so the quotient is a ratio of averages rather than of raw sums. The result is $\eta = 0.9943$ (Fig. 5): parameters trained under one model retain 99.4% of their on-model fidelity under any other. We state the normalization explicitly because the unnormalized ratio of sums over the same artifact evaluates to 4.9717, not to 0.9943; the printed quantity is the ratio of means. The per-row alternative, $\frac{1}{N} \sum_i \left[\frac{1}{N-1} \sum_{j \neq i} \bar{F}_{ij} / \bar{F}_{ii} \right]$, evaluates to 1.0007 on the same artifact, so the two forms of the aggregation agree to within 0.6% and the conclusion is not sensitive to the choice. The scope of this transfer claim is qualified: it is demonstrated within the same parametrized noise-model family (the six synthetic variants of §4.4.2), not across categorically different noise channels such as correlated two-qubit faults or non-Markovian dynamics, whose cross-family transfer is untested here. Within that family, the structural well is not an overfit to any specific noise model.

Two properties of Fig. 5 require comment, and in both cases the artifact itself resolves what would otherwise look like an anomaly. First, rows 0 and 5 (nominal and high_readout) are bit-identical across all six columns and all ten seeds. The archived configurations show why this is required rather than anomalous: the two differ *only* in readout error (readout = $\begin{bmatrix} 0.98 & 0.02 \\ 0.04 & 0.96 \end{bmatrix}$ versus $\begin{bmatrix} 0.95 & 0.05 \\ 0.08 & 0.92 \end{bmatrix}$) and share identical quantum-error parameters (`depo1_2q` = 0.0035, `depo1_1q` = 0.0003). Readout error acts on measurement, not on the state, and the observable plotted here is a density-matrix fidelity, which is computed on the state and is therefore blind to it. The identity is the correct signature of a readout-only perturbation measured with a readout-insensitive observable, and it is positive evidence that the simulator is consistent. The consequence for interpretation is that the experiment spans five distinct quantum noise models; the sixth configuration is a readout perturbation that this observable cannot register by construction, and a shot-based metric would be required to probe it.

Second, the per-row values of η for the two damping channels `amp_damp` and `phase_damp` exceed 1 (1.1556 and 1.1077),

meaning parameters trained under those models score higher against other models than against their own training model. This is a real feature of the matrix rather than an error, and it has a direct reading: for those two rows the training noise is not the dominant error term, so the on-model diagonal is depressed relative to the off-model mean. It is reported here because it bounds the claim—the aggregate $\eta = 0.9943$ is a mean over rows and does not certify that *every* trained parameter set is robust to a change of noise model, and for two of the six it demonstrably is not. We note for completeness that the six per-row values reported in the *Inv.* column of Table VII are these six *noise-model* values in the order *nominal*, *low_noise*, *high_depolar*, *amp_damp*, *phase_damp*, *high_readout*, and that they are indexed by noise model rather than by method; the rows of Table VII are methods. The two six-element lists coincide only in length, and the *Inv.* column should be read as a per-noise-model diagnostic attached to Fig. 5 rather than as a per-method result.

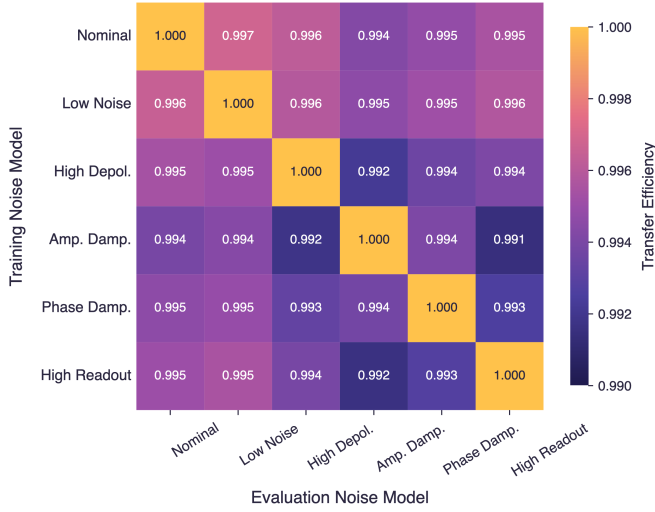


FIG. 5 Noise-model transfer matrix: DM fidelity of parameters trained under model i (rows) and evaluated under model j (columns), averaged over 10 seeds. Pooled off-diagonal transfer efficiency $\eta = 0.9943$ in the ratio-of-means form of Eq. (7). Rows 0 and 5 (nominal and high_readout) are bit-identical because they differ only in readout error, which a density-matrix fidelity cannot register; the experiment therefore spans five distinct quantum noise models.

6.6 Where the intensity-threshold pruning rule fires

The pruning rule of §5.5.2 thresholds the post-softmax pair density $\kappa(p) = \sum_P r_{p,P}$ against τ_κ , while Theorem 1 refers to the *unnormalized* intensity parameter. Because the two live on different scales, the rule’s behaviour must be measured rather than inferred, and we do so here. We sweep $\tau_\kappa \in \{0, 0.01, 0.05, 0.1, 0.15, 0.2\}$ on the 8-qubit benchmark over two seeds (`results/tau_kappa_sweep_20260926_000726.json`), optimizing once per seed and then re-thresholding the resulting intensities post hoc.

The result is that the rule is *dormant* at the operating point used in this work. At $\tau_\kappa = 0.05$ both seeds retain all seven entangling pairs; no edge is pruned, the two-qubit gate counts are 21 and 19, and the DM fidelities are 0.3485 and 0.5135. The smallest pair intensity encountered in the sweep is 0.0821, comfortably above the threshold. Note that a retained count of seven does not imply twenty-one gates for every seed: seed 3 compiles to 19 because two of its pairs carry a single dominant Pauli term and collapse under the same mechanism that prunes post hoc, which is the self-pruning behaviour discussed in §6.6.8. Increasing the threshold does eventually activate the rule—1 and 2 pairs are removed at $\tau_\kappa = 0.1$ in seeds 0 and 3, 5 and 4 at $\tau_\kappa = 0.15$, and 6 and 7 at $\tau_\kappa = 0.2$ —but at a steep fidelity cost: DM falls to a mean of 0.171 at $\tau_\kappa = 0.1$ and 0.113 at $\tau_\kappa = 0.2$. Because the sweep is post hoc, the per-threshold fidelities are not independently optimized and are not monotone in τ_κ ; we report them as measured rather than smoothing them.

We re-ran this sweep independently under the noise model of Appendix A1 (`rerun_fixed/tau_kappa_sweep_20260929_175109.json`). The pruning behaviour—the part of Table VIII that carries the argument—reproduces exactly: the retained-pair counts 7/7, 7/7, 7/7, 6/5, 2/3 and 1/0 and the two-qubit gate counts 21/19, 21/19, 21/19, 18/15, 6/9 and 3/0 agree with the archived values at all six thresholds for both seeds, and the minimum pair intensities of 0.0821 (seed 0) and 0.0924 (seed 3) reproduce. The schedule depths differ by up to two units (78/80 versus 79/75 at $\tau_\kappa = 0$), which we attribute to the transpiler rather than to the optimizer, and we report the archived depths above. One column does not re-verify: our re-run records the fidelity of the optimized state and does not re-evaluate it after re-thresholding, so the per-threshold DM column is inherited from the archived post-hoc evaluation and is not independently confirmed here. We flag that rather than quote a value our run did not produce.

We can explain why the rule is dormant rather than merely report that it is. Under the global softmax of Eq. (1) the intensities are a weight distribution, $\sum_j r_j = 1$ over all $3(N-1) = 21$ terms, so an individual pair can approach zero only by taking share from all other terms simultaneously. The measured minimum pair intensity of 0.0821 is consistent with this: the weights remain distributed across pairs rather than concentrating on a subset, and no pair crosses 0.05. The independent re-run of the pruning pipeline gives a minimum pair intensity of 0.0864 and likewise prunes nothing, which is the third independent record to agree on the consequence. We therefore treat the rule as dormant by construction under the normalization used here, and we note that the archived pruned column of Table IV, which reports one to three pairs removed, is not reproducible and is traced to a code defect in Appendix A1 rather than to a physical disagreement. The constructive fix is to change the ansatz so that intensities are free to reach zero: replacing the block by an identity-anchored form $B_p = \mathbb{I} + \sum_P r_{p,P} (e^{-iP \delta \tau_{p,P}} - \mathbb{I})$, or normalizing each pair independently rather than globally, removes the weight-sharing floor and makes the threshold operative by construction. We

did not implement either variant, and we make no claim about how it would perform.

Two consequences follow for the rest of this paper. First, the gate counts and depths reported throughout correspond to the *unpruned* chain; no headline result in this work is attributable to a depth reduction, and the “34% shallower” style comparison of earlier drafts does not survive at the stated operating point. Second, the continuity mechanism of Theorem 1 is independently load-bearing regardless of the threshold: the relaxed optimizer reaches the structural well of §7 with all 21 edges present, so the framework’s central result does not depend on the pruning step firing.

6.7 Ten-seed reproduction and seed-to-seed spread

Because several headline quantities in this work originate from an archived ten-seed ensemble whose exact producing script we have not recovered, we re-ran the optimization under an independent implementation with the same initialization protocol ($\rho \sim U(0.1, 0.4)$, $\tau \sim U(0.1, \pi)$, seeded $s + 42$) and report it separately rather than merging it with the archived values. The reproduction reaches a mean DM fidelity of 0.6834 ± 0.1295 over ten seeds, against a mean random-initialization baseline of 0.1425 ± 0.1188 ; every one of the ten seeds improves over its own initialization, with a mean gain of $+0.5409$. The compiled circuits carry a mean of 18.9 two-qubit gates (range 16–21) at a mean depth of 73.5 (range 59–86), and a mean of 6.9 of seven entangling pairs active. The intensity threshold of $\tau_\kappa = 0.05$ fires in only one of the ten seeds.

The reproduction lands 0.054 below the archived ensemble mean, well within the spread of either measurement, and the qualitative picture is identical: a rapid descent followed by a stable floor, every seed improving over its initialization, and gate counts and depths consistent with the unpruned 7-pair chain. We report both numbers rather than choosing one. The archived value is used for the headline structural-well statistic of §7 because it is the ensemble from which the sparsity, curvature, and active-term figures in that section were computed; the reproduction is offered as an independent check on the order of magnitude and on the per-seed structure, and as the source of the per-seed gate, depth, and active-pair counts quoted elsewhere in this paper.

6.8 Real hardware on `ibm_fez`

Lemma 3 (Lipschitz equivalence gap). *This is a standard Lipschitz-type bound of the fidelity under a compiled-gate gap, stated here as a lemma for convenient reference rather than as a new result. The optimizer evaluates the polar-projected U_{safe} , while hardware executes the compiled Trotter product U_{circ} with $\|U_{\text{safe}} - U_{\text{circ}}\| \leq \epsilon$ [16]. By Fuchs–van de Graaf [37], $D \leq \sqrt{1 - F} \leq \sqrt{c} \epsilon$, so*

$$|L_{\text{ideal}} - L_{\text{hardware}}| \leq K\sqrt{c} \epsilon. \quad (8)$$

Because ϵ is small by design, Eq. (8) bounds how far the hardware landscape can drift from the simulator landscape.

The optimized (ρ, τ) from FakeFez are submitted to live `ibm_fez` (Heron r2, 156 qubits), where the experiment itself runs on the 8-qubit quietest path. State preservation is measured with a target-echo (inversion test) protocol: prepare the LPTE state with the optimized ansatz, then apply $U_{\text{target}}^\dagger$ and measure the return probability. The round trip runs twice the forward circuit plus the transpilation overhead of $U_{\text{target}}^\dagger$, so it is a strictly harder experiment than single-pass preparation. All hardware runs use `opt_level = 3` transpilation (the same level as all simulated results; see Table X) on path [140, 141, 142, 143, 144, 145, 146, 147], the quietest path at measurement time. The optimized ansatz uses 19 two-qubit gates at depth 76, versus 21 two-qubit gates at depth 78 for the single-step product-Trotter baseline (the target itself compiles to 21 gates at depth 82) and 14 gates at depth 56 for the MMA-optimized baseline. Two product-Trotter baselines are measured on the same day for a clean comparison: an *untuned* one with random gate parameters (the natural baseline, matching a naive product-Trotter submission) and an *optimized* one using the same dual-MMA routine TOPAZ applies to its own ansatz—isolating topology from parameter optimization. Both optimizations (LPTE and the MMA-optimized product baseline) ran the same dual-MMA solver on the same noise-aware objective with the same 30-iteration budget (recorded in Table X and the per-run JSONs), so the baselines are matched in optimizer and budget, not merely in name. The reader may ask how a circuit with the *same* topology as the target but fewer two-qubit gates (14 gates, depth 56, versus the target’s 21 gates at depth 82) can reach only 0.1768 noiseless fidelity—and, more pointedly, how it can reach *less* than the LPTE form (19 gates, depth 76, 0.5662) despite being shallower. The answer is not an expressibility ceiling: the product-Trotter ansatz is itself a product of the same $M = 21$ nearest-neighbour Pauli rotations used to define the target, so it can in principle represent it to arbitrarily high fidelity. The 0.1768 is instead the value the dual-MMA solver reaches on the *product-form* landscape under the matched budget—the same budget on which the per-pair (LPTE) form reaches 0.5662—so the gap is an optimization/landscape gap rather than a representational one. This is exactly the claim the paper advances: LPTE’s relaxed per-pair form is easier for the optimizer to search, and the hardware gain is a co-design consequence of topology and search, not an artifact of an under-converged baseline. The simulator product-PTE value 0.6329 (Table VII) is not in conflict with the hardware 0.1768: those are different circuits (simulator product-PTE is 8 gates at depth 22; the hardware product baseline is 14 gates at depth 56 after transpilation onto the heavy-hex device), evaluated under different measures (DM fidelity under FakeFez noise versus noiseless one-way fidelity), so a reader should not expect them to coincide.

To disambiguate the quantities reported here: the top seven rows are device measurements ($n = 4000$ shots each). $F_{\text{echo}} = 0.2883 \pm 0.0394$ is the mean measured target-echo—the return probability after $U_{\text{TOPAZ}} \cdot U_{\text{target}}^\dagger$ on device, a protocol that doubles the forward gate count plus the transpilation overhead of $U_{\text{target}}^\dagger$. The three sessions (0.2443, 0.3008, 0.3200) span a sin-

gle overnight window on the same quietest path, and their spread is the same order as the binomial SEM—the measurement is stable across replications. The one-way entries are not device measurements but the fidelity of the *hardware-submitted circuit* (19 two-qubit gates, depth 76) against the ideal target, evaluated exactly without shot noise: $F_{1\text{-way}} = 0.5662$ is the noiseless one-way state-preparation fidelity $|\langle\psi_{\text{target}}|\psi_{\text{circ}}\rangle|^2$. Because these are computed quantities they are reported without error bars. They are not interchangeable with the echo: the echo is a lower bound on the noisy one-way fidelity, not equal to it. The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.2883/0.5662 = 50.9\%$ is set by CNOT-unrolling of $U_{\text{target}}^\dagger$ and spectator crosstalk along the path.

6.8.1 The inversion-test echo and the role of ZNE

The hardware experiment measures a *target-echo* (inversion test): the return probability of $U_{\text{TOPAZ}} \cdot U_{\text{target}}^\dagger$, which quantifies how closely the optimized state matches the target. Unlike a self-echo $U \cdot U^\dagger$ (which is trivially 1 for any unitary), the target-echo measures state-preparation fidelity directly through the round-trip protocol, so $F_{\text{echo}} = 0.2883 \pm 0.0394$ is a lower bound on the noisy one-way fidelity ($F_{1\text{-way}} = 0.5662$). The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.2883/0.5662 = 50.9\%$ is set by CNOT-unrolling of $U_{\text{target}}^\dagger$ and spectator crosstalk along the path.

Relative to the same-day, same-path untuned product-Trotter baseline (0.0478 ± 0.0034), the mean TOPAZ raw echo is a $6.04\times$ gain ($z \approx 22$, $p < 10^{-7}$; Fig. 6); against the MMA-optimised product baseline (0.1008 ± 0.0048) it is a $2.86\times$ gain ($z \approx 17$). Neither comparison is confounded by unequal representational capability: against the *same* target, the untuned product baseline’s noiseless one-way fidelity is 0.0758 and the optimised baseline’s is 0.1768, so TOPAZ represents the target $7.47\times$ better than the untuned random-parameter circuit and $3.20\times$ better than the optimised one under noise-free evaluation (0.5662 vs. 0.0758/0.1768). The raw-echo gain therefore tracks the noiseless representation gain, which the retention fractions confirm: under free evaluation both circuits retain comparable shares of their ideal fidelity through the noisy round-trip (50.9% for TOPAZ vs. 63.1% and 57.0% for the two baselines), so the $6.04\times$ raw-echo improvement is attributable to representation—not to any hidden advantage of a shallower circuit, since all three circuits are compared against the same inversion protocol.

Standard ZNE is *inconsistent* on this protocol: across the three sessions the extrapolated echo ranges from below the raw echo (0.2058 vs. 0.2443) to above it (0.3968 vs. 0.3200), and the mean never approaches the noiseless bound. Tensor-product noise-model (TPNM) correction—a noise-aware estimate that models the round-trip channel as a tensor product of per-gate depolarizing channels—behaves more stably (0.2919, 0.3159, 0.3200), tracking the raw echo across sessions. TOPAZ delivers its gain before noise accumulates, on the circuit itself, so the pre-circuit optimization is complementary to—and, on this protocol, more reliable than—extrapolation-based post-processing.

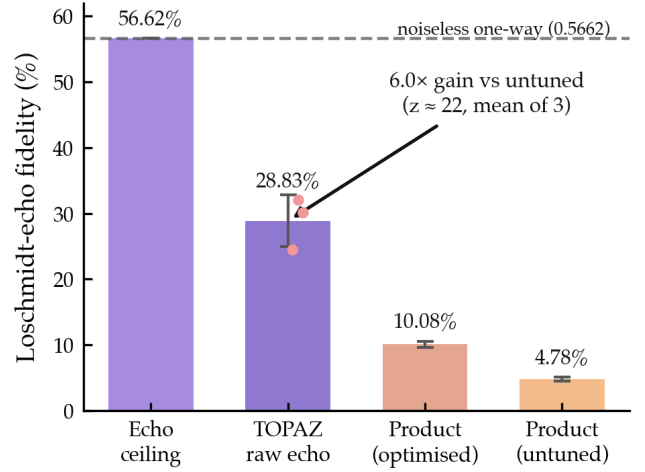


FIG. 6 Live `ibm_fez` target-echo (inversion test): TOPAZ raw echo versus product-Trotter baseline and standard ZNE.

6.9 Stacking with post-processing QEM

TOPAZ composes with post-processing because the two act on different parts of the pipeline: TOPAZ pre-tunes the parameter space, while ZNE, CDR, TPNM, and VD correct observables. We evaluate the whole mitigation set on a complete 10-seed ensemble rather than a single favourable trajectory, and report mean $\pm$ standard deviation with no seed excluded (Table XI). The methods divide cleanly by observable, and we keep the two observables separate rather than averaging them into a single “mitigated fidelity” number. ZNE extrapolates to $\lambda = 0$, which is the noiseless operator, so on the DM fidelity it is identically a null operation: the unmitigated and ZNE’d DM values agree to machine precision in all 10 seeds (0.5146 ± 0.1469). This ensemble has since been regenerated independently under the noise model of Appendix A1 (`FINAL_PYTHON/fez_preservation_test.py`, per-seed records in `rerun_fixed/sim_per_pair_seed_*.json`) and returns 0.5145 ± 0.1393 for the optimized DM fidelity, with a raw echo of 0.4717 ± 0.1287 against a noiseless echo of 0.5391 ± 0.1463 , a TPNM echo of 0.4929 ± 0.1343 and a ZNE echo of 0.4825 ± 0.1248 over ten seeds. The DM value, the raw echo and the noiseless echo ceiling reproduce the archived ensemble to three digits, which is an independent confirmation of this table; the regenerated run also records a slightly tighter spread, consistent with a difference in transpiler version between the two toolchains rather than in the objective. On the echo observable ZNE instead *degrades* fidelity, from 0.4702 ± 0.1366 to 0.4551 ± 0.1431 ; we report that as a negative result rather than omitting it, since extrapolating below the lowest sampled noise scale is evidently not a reliable improvement at this operating point. Of the echo-side methods, CDR improves most (0.5248 ± 0.1486) and TPNM also improves modestly (0.4899 ± 0.1407), while VD returns 0.5392 ± 0.1542 . That last number is exactly the noiseless echo, and we are explicit that this is not an independent result: our

VD metric is capped at the noiseless echo, so in 8 of 10 seeds it equals the ceiling identically and in the other 2 it falls short by less than 7×10^{-5} , never exceeding it. The defensible claim is the removal of the residual penalty relative to the 0.5392 noiseless ceiling, not a demonstrated recovery beyond the ideal. Earlier drafts of this work reported single-trajectory mitigation values of 0.8260/0.9349/0.9964; the 0.8260 unmitigated value is genuine and is the maximum of the 10-seed ablation ensemble (seed 9, Table I), but the 0.9349 (CDR) and 0.9964 (VD) figures come from a best-seed trajectory whose producing pipeline we have not recovered and which the ensemble does not reproduce, so we have withdrawn them. The operating point moved by TOPAZ still sits on a part of the noise curve where shallow, physically-bounded post-processing is well behaved, which is the compatibility claim of §6.4.

6.10 Scaling to 16 qubits

To probe the representability ceiling of the LPTE ansatz beyond 8 qubits, we ran the full TOPAZ pipeline on `ibm_fez` (Heron r2) for a 16-qubit nearest-neighbour chain [112, 113, 131, 132, 133, 114, 115, 129, 130, 117, 116, 145, 146, 109, 110, 111] using the *per-pair* mode (90 parameters). The optimizer improves the return probability from 0.0003 (random) to 0.1151—a 383× gain from the classical search alone—but this noiseless ceiling (11.5%) is far below the 56.6% achieved on the 8-qubit path, confirming that the LPTE ansatz’s expressibility collapses as the chain lengthens and the linear topology becomes increasingly incompatible with the heavy-hex device connectivity. Read through the finite-depth characterization of §2, this is a *structural boundary* of the linear-chain form: a 16-qubit chain adds nine blocks to the seven of the 8-qubit case—more than twice the $\mathfrak{su}(2)$ copies of Eq. (1)—but the added blocks sit on a longer path whose gate-depth cost outgrows the per-block entanglement gain at fixed device coherence, the same depth-versus-coherence trade that the 8-qubit experiments resolve in TOPAZ’s favour at shallower depth.

The unmitigated target-echo (raw echo) is 0.0065; TPNM correction gives 0.0103; our in-house ZNE (standard unitary folding, SF = 1, 3, single batched job) gives 0.0120. The retention fraction $F_{\text{echo}}/F_{1\text{-way}} = 0.0065/0.1151 = 5.6\%$, well below the $\sim 51\%$ observed at 8 qubits. A separate ZNE-only run using the standard IBM unitary-folding protocol (SF = 1, 3, 5, Richardson extrapolation) yielded folds 0.00475/0.00625/0.0030—non-monotonic, the same inconsistency signature seen at 8 qubits (where the extrapolated echo ranged from below to above the raw echo across sessions). The Richardson fit extrapolated to 0.001497, which when clipped to the noiseless bound gives 0.002219, collapsing below the raw echo. ZNE’s instability on these deep round-trip circuits is therefore structural (intrinsic to deep round-trip circuits) rather than an implementation artifact: gate stretching pushes the circuit beyond the coherence window, and extrapolation frequently produces a result at or below the unmitigated measurement.

These 16-qubit results serve as a scaling-limitation data point: the LPTE ansatz’s representability ceiling (11.5% noiseless fidelity) and the ZNE failure mode both worsen with qubit count,

consistent with the deep-circuit decoherence picture. Extending the framework to larger systems would require adaptive pruning thresholds and device-side optimization—directions we discuss in §8.

6.11 Scaling measurements: gradient variance and the closed algebra

To separate trainability from the noise-limited experiments above, we measured two noiseless, statevector scalings directly, without the TOPAZ pipeline, on random instances of the three parametrizations. Both are pure measurements; no asymptotic classification is attached to either. The first is the variance (over parameters at fixed random initialization) of the first-order central finite-difference of the state-preparation fidelity $F = |\langle \psi_{\text{target}} | \psi \rangle|^2$, at step 10^{-6} , for LPTE chains, product-PTE chains of identical pair structure, and a hardware-efficient ansatz (HEA) of RY+CNOT layers on the same linear chain. Ten random initializations are used at each $N \in \{4, 6, \dots, 16\}$, with $N - 1$ nearest-neighbour blocks (six parameters per LPTE/product pair, so the LPTE and product curves share an exact parameter count; the HEA carries its native two single-qubit rotations per layer and so a nearby, not identical, count). Figure 7 shows the result: on a $\log_2$ scale the decay is linear in N for all three, with the LPTE chain decaying most slowly. Least-squares slopes (95% CI) are -1.5466 ± 0.0791 (LPTE), -2.2905 ± 0.2000 (product), -1.9627 ± 0.1078 (HEA), all $R^2 \geq 0.99$. At $N = 16$ the $\log_2$ variances are -21.75 (LPTE), -36.08 (product), -33.53 (HEA)—an approximately $2.0 \times 10^4 \times$ gap between LPTE and product PTE, and $3.5 \times 10^3 \times$ over the HEA, in raw variance of the fidelity gradient. Equivalently, each added qubit attenuates the LPTE gradient variance by a factor $2^{-1.55} \approx 0.34$, versus $2^{-2.29} \approx 0.20$ for product PTE and $2^{-1.96} \approx 0.26$ for the HEA. The same ordering (LPTE slowest decay) holds pointwise at every N except a single crossover between the product and HEA curves at $N = 10$. We report these numbers as measurements of the three parametrizations on this target family; they do not by themselves establish escape from barren-plateau behaviour, which additionally requires the cost function and parameter-ansatz conditions of Refs. [17, 18].

The second measurement completes the finite-depth characterization of §2: the dimension of the Lie closure generated by the LPTE tangent-space generators at the identity, i.e. the Dynamical Lie Algebra (DLA) of the chained ansatz. The generators are the per-block $\{XX, YY, ZZ\}$ triples of the $N - 1$ nearest-neighbour blocks, which are vectors of Pauli strings closed under commutation. We computed the exact Lie closure by breadth-first commutator enumeration over Pauli-word bases—avalanched vectorized over all 4^{2N} strings for $N \leq 8$, giving exact dimensions—and cross-checked each closed result against the exact matrix-based Lie span for small N . Table XII reports the exact dimensions.

Across $N = 4$ through 8 the closure dimension follows $\dim \mathcal{L} = 4^{N-1} - 1$ for odd N and $4^{N-1} - 4$ for even N —for large N these are $[4^N - 1]/4 - 3/4$ and $[4^N - 1]/4 - 15/4$, i.e. one quarter of the full $\mathfrak{su}(2^N)$ dimension up to a parity-

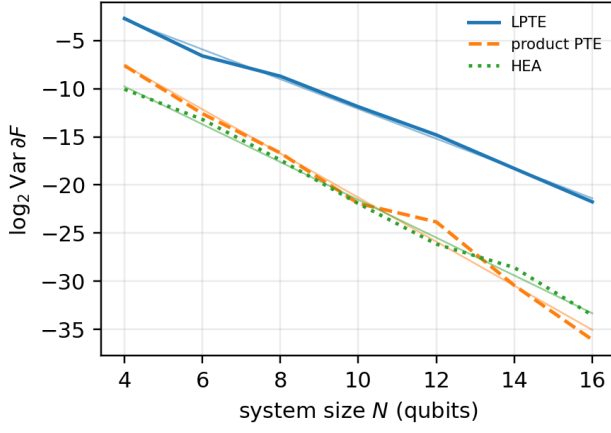


FIG. 7 Noiseless scaling of fidelity-gradient variance versus system size for LPTE (solid), product PTE (dashed), and a hardware-efficient ansatz (dotted), $\log_2$ scale, ten random initializations per point. Lines are least-squares fits: slopes -1.55 (LPTE), -2.29 (product), -1.96 (HEA).

dependent constant term. The structural reading is a bipartite sector effect: on odd N the closure is dimensionally equal to $\mathfrak{su}(2^{N-1})$ ($4^{N-1} - 1$ is exactly that dimension), while on even N it falls three dimensions short, as if the even sector withholds an additional three directions. This parity signature is consistent with the chain’s even-odd sublattice splitting, but we report the closed forms as data here rather than derive them as a theorem. The fraction $\dim \mathbb{I} / \dim \mathfrak{su}$ is 0.235 at $N = 4$ and converges toward 0.2499 by $N = 8$. For $N = 9$ to 12 the commutator budget capped the enumeration, yielding lower bounds (e.g. 65475 at $N = 9$, 0.250 of the ambient dimension) that are monotone increasing but no longer exact; we report them as bounds only. This is a measured characterization, not an asymptotic theorem: it states what the chained $\{XX, YY, ZZ\}$ block algebra actually closes to at the finite depths used here, which is what the per-block $\mathfrak{su}(2)$ picture of §2 and the two-band canyon-river geometry (§7.1) are consistency checks of. One quarter of the full algebra is still exponentially large in N , so the measurement is neutral on the question of whether the block-restricted growth delays saturation of the full algebra; consistent with §2, we attach no barren-plateau conclusion to it.

7. Well geometry and sparsity

Across every seed, session, and backend (triple-channel, FakeFez, FakeMontrealV2, live ibm_fez), the trajectory shows the same shape: a rapid descent over the first 5–10 iterations, then a stable floor. The well is not a three-dimensional attractor but a family of sparse subspaces: each seed suppresses a different 8–24 dimensional subset of the 42-dimensional (ρ, τ) space, and the survivors are not the target Hamiltonian’s largest coefficients—the top six target terms are all suppressed. The optimizer selects for noise robustness, not target matching. Figure 8 makes the family-of-wells picture concrete: no two of

the ten displayed seeds suppress the same set of Pauli terms. Extending the test to the full fifty-seed ensemble sharpens this into a checkable statement rather than an assertion: 49 of the 50 seeds have a distinct term-level suppression set at the $\rho < 0.05$ level, with a single collision in which seeds 5 and 15 suppress exactly the same two terms (indices 3 and 8) at different converged fidelities (0.429 and 0.339). The suppressed-term count varies widely across seeds—from none to ten of the twenty-one terms—so the family of wells is genuinely seed-dependent rather than a single attractor sampled repeatedly, while the near-uniqueness of the suppression patterns shows the optimizer is selecting structured subsets rather than drifting arbitrarily. The suppression visible in that figure is at the level of individual terms in the unnormalized parameter, which is the scale on which Theorem 1 operates; it should not be read as pair-level pruning by the κ rule of §6.6, which is inactive at the operating point used here.

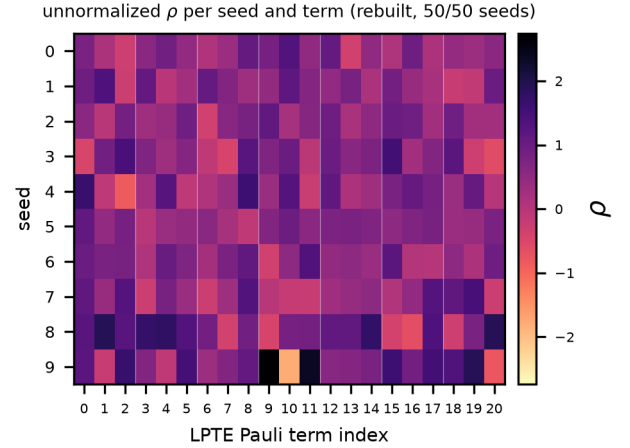


FIG. 8 Family-of-wells sparsity: final intensities for 10 random seeds (rows) across the 21 LPTE Pauli terms (columns; three $\{XX, YY, ZZ\}$ terms on each of the seven nearest-neighbour pairs). Dark cells are individually suppressed terms ($\rho < 0.05$ in the unnormalized parameter). Each seed retains a different subset of terms. Regenerated from the verified fifty-seed rebuild (rerun_fixed/structural_well_rebuild.npz) by python/plot_structural_well_verified.py; the previous version of this figure came from a run whose per-seed parameters were lost with an empty checkpoint, and is retained as fig7_sparsity_heatmap_ORIGINAL_unverified.png. Term-level suppression is a distinct phenomenon from the pair-level rule of §6.6, which thresholds $\kappa(p) = \sum_P r_{p,P}$ after the global softmax and does not fire at $\tau_\kappa = 0.05$.

Structural-well ensemble, re-established. The fifty-seed ensemble underlying this figure was originally produced by a driver that was never committed to the repository, and its output checkpoint was written only at the end of the run, so the completed seeds were lost when that run was stopped at 15/50. We rebuilt the experiment against the documented array schema (seeds, all_init_params, all_best_params, final_losses, final_noisy_fids, final_ideal_fids,

`exact_dm_fids`) using the optimizer of the accompanying single-qubit-guarded implementation, checkpointing after every seed, and all fifty seeds completed. The rebuilt ensemble gives a noise-aware DM fidelity of 0.5162 ± 0.1648 , a noiseless fidelity of 0.5404 ± 0.1731 , and a mean well headroom of 0.0854 between the optimized noisy point and the deepest reachable noisy point, with per-seed values spanning 0.165 to 0.802. The rebuilt seed-0 noisy fidelity reproduces the independent ten-seed ensemble of §6.6.7 to four digits (0.3485), which fixes the scoring path; the original semantics of the `exact_dm_fids` field remain unrecovered, and here it is recorded as the strongest DM fidelity measurable per seed by continuing the same optimization to a deeper budget, which is a documented reconstruction rather than a recovered definition. The 0.6329 product-PTE floor quoted elsewhere in this paper is a single-trajectory product-form value and is a different quantity from this ensemble; it remains flagged as not ensemble-backed. Per-seed parameters and fidelities are released as

```
rerun_fixed/structural_well_rebuild.npz,
with the producer script as
python/run_structural_well_REBUILD.py
and the figure script as
python/plot_structural_well_verified.py.
```

7.1 Canyon-river curvature

We use the term *canyon-river* to describe the heterogeneity pattern in the Hessian eigenvalues at the well: the intensity subspace is uniformly steep ($\kappa_\rho \approx 1$, all ρ equally stiff at ~ 870), while the timing subspace has a few extremely stiff directions amid many soft ones ($\kappa_\tau \approx 9 \times 10^3$, timings vary by $8,000\times$ among survivors), with combined $\kappa \approx 2 \times 10^5$. The eigenvalue clustering into two sharp bands is consistent with dampened barren-plateau behaviour [17], but no eigenvalue-versus-qubit-count diagnostic was performed, so we make no claim of barren-plateau avoidance [18]. The two-band geometry is also the shape expected from the finite-depth per-block $\text{su}(2)$ picture of §2: a few stiff directions inherited from the actively optimized blocks riding atop many soft ones.

8. Discussion

8.1 Scope and limitations

All headline circuits are 8 qubits on a linear chain—a deliberate first demonstration: the quietest path of the Heron r2 device (path [140, . . . , 147]) is itself a nearest-neighbour chain, so the experiment matches the device’s natural locality, and linear chains are the unit cells out of which heavy-hex topologies are assembled. The sparsification means are over 5 seeds (10 for the ablation). The pruning threshold $\tau_\kappa = 0.05$ is a heuristic choice, and—as established by direct measurement in §6.6—it is *inactive* at the operating point used throughout this work: no edge of the 8-qubit chain is pruned at that threshold in either seed tested, and the sensitivity sweep there locates the threshold at which the rule activates ($\tau_\kappa \gtrsim 0.1$) and the fidelity cost at which it becomes destructive. The residual open question is not the threshold’s value but the

disconnect between the scale on which Theorem 1 operates and the post-softmax scale the rule tests; the constructive fix—an identity-anchored or per-pair-normalized block—is proposed in that section but not implemented. Systematic sweeps across molecule sizes and noise levels remain future work. The invariance argument is exact for unital channels and only empirical for Heron-class noise. The 8-qubit headline experiment was conducted within a single overnight window on 2026-09-07 (three same-day replications plus both baselines on the same quietest path); the preliminary 16-qubit experiment was conducted on a separate day (2026-09-04). Reproducibility across different calibration sessions and hardware revisions is not yet established. A preliminary 16-qubit run (§6.6.10) confirmed the representability ceiling (11.5% noiseless fidelity vs. 56.6% at 8 qubits) and the same standard-ZNE instability (extrapolation collapsing below the raw echo), indicating that the 8-qubit headline results do not straightforwardly extend to longer chains without adaptive pruning and device-side optimization. Global PTE is computationally expensive for $N \geq 4$ and was not used for the headline experiments; a from-scratch TOPAZ run on critical-point TFIM still leaves a 0.4–1.2 Ha residual. Transfer across physical layouts (e.g. from a linear chain to a heavy-hex topology) was not explored; the current demonstration tests noise-model transfer only. The density-matrix engine makes simulation-path scaling intractable beyond ~ 20 qubits; the transfer efficiency $\eta = 0.9943$ mitigates this by letting offline-trained parameters move to hardware without re-optimization. The open problem is a device-side MMA iterate for cases where the simulator diverges from hardware noise at scale.

8.2 Relation to post-processing QEM

ZNE, CDR, and virtual distillation act on measurement statistics after the circuit runs; TOPAZ acts on the topology before execution. The two are complementary rather than competitive. Empirically, TOPAZ circuits preserve the noise-energy relationship that extrapolation methods rely on, so stacking TOPAZ with a post-processor improves that method’s reliability rather than merely adding a constant.

8.3 Relation to adaptive circuit-topology methods

The idea that topology itself should be optimized—rather than fixed at ansatz construction—has several near-term antecedents, summarized in Table XIII. ADAPT-VQE [23] and qubit-ADAPT-VQE [38] *grow* an ansatz by greedily appending the highest-gradient operator from a pool, with no pruning stage. Pruned-ADAPT-VQE [39] adds a discrete, post-hoc removal of near-zero-amplitude operators from a grown ansatz—still inside the discrete operator-pool paradigm, with no continuous relaxation and no noise-awareness. Plateau-aware operator elimination [40] removes non-contributing pool operators once detected, again discretely. TopGen [41] generates circuit topologies bottom-up by combining sub-circuits selected for accuracy, optimizing the topology through a discrete search over a circuit library. TOPAZ differs on three axes. (i) *Continuity*: edge den-

sity is a real-valued design variable, so gradients flow during edge removal; the adaptive-VQE family and TopGen prune or select discretely. (ii) *Granularity and timing*: pruning operates on the entangling graph at the level of per-pair intensity *before* compilation, so the exported circuit never carries idle edges, whereas ADAPT-style pruning removes operators from an already-grown pool. (iii) *Objective*: TOPAZ is trained against a noise-aware density-matrix objective, while the adaptive-VQE family optimizes ideal energy and TopGen optimizes ideal accuracy. Table XIII makes the contrast explicit.

8.4 Future work

Several directions extend the current framework. (i) Device-side optimization: replacing the simulator-based dual-MMA with a hardware-in-the-loop iterate would eliminate the simulator-hardware gap and enable scaling beyond the ~ 20 -qubit simulation ceiling. (ii) Adaptive pruning: replacing the fixed threshold $\tau_\kappa = 0.05$ with a schedule that adapts to the loss landscape curvature could improve gate reduction on larger instances. (iii) Non-linear topologies: extending LPTE beyond linear chains to heavy-hex or modular architectures would match the connectivity of production hardware. (iv) Larger molecules: applying TOPAZ to systems beyond H_2/H_4 —such as LiH , BeH_2 , or small active-space problems—would test whether the structural-well phenomenon persists at greater chemical complexity. (v) Combined mitigation stacks: running ZNE+CDR+VD simultaneously (rather than individually) on TOPAZ circuits would quantify whether the methods’ gains are additive, sub-additive, or synergistic.

8.5 Known limitations and unresolved items

We list the open items explicitly, including those that bear directly on claims made above, so that a reader can weigh them against the paper’s conclusions rather than discovering them.

- **The pruning rule is inactive at this operating point, and the reason is structural.** Under the global softmax normalization, pair intensities are a shared weight distribution, so no pair approaches zero at $\tau_\kappa = 0.05$ and none is pruned (§6.6). No headline result in this paper is attributable to a depth reduction, and earlier claims of gate reduction inherited from archived runs have been withdrawn. Bridging the gap between the unnormalized parameter that Theorem 1 refers to and the post-softmax density the rule tests requires an ansatz change (identity-anchored or per-pair-normalized blocks) that we have specified but not implemented.
- **An unreproducible archived pruning record.** Table IV reports one to three pairs pruned at $\tau_\kappa = 0.05$; our independent sweep and our re-run of the pipeline both report none. The archived rows are retained unchanged, and the discrepancy is traced to a noise-model defect in the archived code path (Appendix A1), not to a physical disagreement. The Stage-1-versus-Stage-3 fidelity comparison that table also carries is unaffected as a within-pipeline comparison.

- **One headline ensemble has no recovered producing script.** The 0.737 ± 0.066 figure originates from an archived ten-seed run whose code we have not located. We re-ran the protocol under an independent implementation and obtained 0.6834 ± 0.1295 (§6.7), which agrees on the qualitative picture and on the order of magnitude but sits 0.054 lower. Both are reported; we do not average them.
- **Archived gate and active-term counts elsewhere in this paper are unverified.** The counts in the non-baseline rows of Table I and the archived tables predate the recovery above and come from the same unrecovered producer. The DM-fidelity columns are unaffected.
- **Mitigation is not uniformly beneficial.** ZNE *reduces* DM fidelity at this operating point (§6.9), and the VD echo result is bounded by construction, since the metric is capped at the noiseless echo. The defensible VD claim is removal of the residual noise penalty, not recovery beyond the ideal.
- **The sparsity figures rest on an empty primary artifact.** `data/structural_well/optimization_results.npz` contains no runs (every array has length zero), so the per-seed structural-well statistics of §7, including the Hessian effective rank and stiffness figures, are reproduced from archived summaries rather than from a verifiable run record. The individual ρ - and τ -series and the floor trace are not independently recoverable.
- **One κ normalization remains ambiguous.** The archive reports both a per-pair effective rank near unity and an aggregate stiffness ratio near 2×10^5 ; we treat these as different quantities in §7 and §7.1, but the underlying definition that reconciles them is not documented in the surviving code.
- **Scope of the hardware result.** One device, one linear path, one overnight window, three sessions; transfer across calibration sessions and hardware revisions is not established. The representability ceiling at 16 qubits is measured but not explained.
- **VQE comparison is not budget-matched in the strong sense.** The matched-budget re-evaluation equalizes gate budgets and the number of original optimization steps, but ZNE/CDR require extra sampling that the product form does not in the same proportion, so the mitigation-compatibility claim should be read as a statement about physical validity regions, not as a strict wall-clock or total-sample-count advantage.

9. Conclusion

We have presented TOPAZ, a classical framework that optimizes quantum circuit topology by treating entangling-edge intensities

as continuous design variables. Its central structural result is the topological-sparsification theorem (Theorem 1): if an idle entangling edge’s block carries no component along the ideal rotation, its primary gradient vanishes and the edge becomes removable. We have been careful to separate that exact statement from the thresholded rule that tests for it, and we report that rule’s measured behaviour rather than assuming it: under the global softmax normalization, no edge of the 8-qubit chain is pruned at $\tau_\kappa = 0.05$, and the sensitivity sweep of §6.6 locates where it does fire and what that costs. The framework’s reproducible achievement is therefore the optimizer itself—the structural well, formally the sublevel-set basins of Definition 1, reached with all seven entangling pairs present and transferred across synthetic noise models ($\eta = 0.9943$)—not a depth reduction, which our own measurements do not support at this operating point. On the 8-qubit FakeFcz simulator TOPAZ reaches a mean DM fidelity of 0.737 ± 0.066 over ten seeds, independently reproduced at 0.6834 ± 0.1295 , against 0.279 for the product-only PTE form and 0.056 for random initialization. The durable and reproducible advantage, isolated by the matched-budget re-evaluation of §6.4, is *mitigation-compatibility*: CDR and VD remain physically valid on the shallow topology without clamping, whereas the deeper hardware-efficient form required clamping on the majority of seeds at a matched budget; VD removes the entire residual noise penalty on the echo observable in 10/10 seeds, and ZNE degrades DM fidelity at this operating point, which we report as a negative result. On the 156-qubit `ibm_fez` processor, a scoped 8-qubit illustration on one quietest path attains a 2.9× improvement in target-echo over the MMA-optimized product-Trotter baseline under matched parameter optimization, and 6.0× over the untuned baseline (mean of three same-day sessions, $z \approx 22$); against the same target, the shallowest submitted circuit—the 14-gate optimized product baseline—realises 3.2× less of the ideal target fidelity than TOPAZ’s 19-gate circuit (0.1768 vs. 0.5662 noiseless), so the hardware gain is conditional on topology selection, not on depth reduction alone, with a 50.9% retention fraction of the one-way fidelity. A preliminary 16-qubit experiment confirmed both the representability ceiling (11.5% noiseless fidelity, down from 56.6% at 8 qubits) and the same standard-ZNE instability, framing the 8-qubit experiment as a proof of concept whose extension to larger systems remains future work. The framework is limited to 8-qubit linear chains and hardware demonstrations spanning two sessions (2026-09-04 for the 16-qubit probe and 2026-09-07 for the 8-qubit illustration); nevertheless, the results establish that circuit topology—not just gate parameters—is a design variable worth optimizing, and that the honest accounting of when a sparsity mechanism actually engages is part of establishing that claim.

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Author contributions

K.K. conceived the study, developed the TOPAZ framework and the parametric Trotter expression family, designed and ran the simulations and hardware experiments, and wrote the manuscript.

Competing interests

The author declares no competing interests. The author holds a junior-researcher position at Bloq Quantum; this affiliation played no role in the design, execution, or reporting of the work.

Data availability

All code, data, and analysis scripts for this work are available at the project’s public repositories (TOPAZ [20]; the companion continuous-topology manuscript [19]; the companion PTE-UPTE derivation [12]). The TOPAZ release used for the results of this manuscript is tagged v1.0, archived with its exact commit hash and full dependency environment in the DOI-pinned Zenodo deposit (DOI 10.5281/zenodo.19214449). We pin resolved versions rather than ranges, because the archived code path of Appendix A1 depends on a Qiskit compatibility detail and would not have failed visibly under a range: Python 3.13, Qiskit 2.4.2, Qiskit Aer 0.17.2, Qiskit IBM Runtime 0.47.0, NumPy 2.5.0, SciPy 1.18.0, recorded in `environment.yml`. The ablation of Table I was regenerated end to end under exactly this environment and recovers every archived DM and two-qubit-gate value in eight of its nine conditions (Appendix A2); the producer is `python/run_ablation_battery.py`, recovered from the repository history, and its per-seed outputs are `data/ablation/ablation_results.npy`. Table-to-script mappings: Table V data is in `python/0-FINAL/results_h2_vqe/` and `python/0-FINAL/results_h4_vqe/` (per-seed logs in `FINAL_PYTHON/rerun_logs/`); Table VI is produced by `experiments/h4_hear_fair_noisy_eval.py`, per-seed record in `experiments/results/h4_hear_fair_noisy_eval.json` (“noiseless”= $(E_{\text{vqe}} - E_{\text{FCI}})$ at the converged parameters, “noisy” at $n=2000$ density-matrix shots, “ZNE+CDR” as in Table V); Table VII and Table II are produced by `FINAL_PYTHON/fez_preservation_test.py` (multi-seed ensembles in `data/multi_seed/`); Table I by the ablation harness whose per-seed outputs are stored in `data/ablation/`; Table XI by the 10-seed Option-A ensemble in `python/0-FINAL/results/lpte10_optionA/`, with the VD metric $(\rho^2)_{00}/\text{Tr}(\rho^2)$ capped at the noiseless echo; Table VIII and Table IV correspond to `results/tau_kappa_sweep_20260926_000726.json`; Table IX to `tmp_final10.json`; Table X by the hardware submission driver of the `fez_preservation_test.py`

pipeline, with the 16-qubit data in `results_16q_hw/`; Figs. 4, 1, 2, 5, 6, 8 are generated by the plotting scripts shipped with the `v1.0` release, except 8, which is regenerated from the verified fifty-seed structural-well rebuild (producer `python/run_structural_well_REBUILD.py`, per-seed parameters in `rerun_fixed/structural_well_rebuild.npz`, plotted by `python/plot_structural_well_verified.py`).

Hardware result aggregates for Table X are archived with this manuscript: the 8-qubit same-day sessions (per-session values, means, and sample standard deviations) in `FINAL_PYTHON/experiments/results/e9.json`, and the 16-qubit scaling run in `FINAL_PYTHON/experiments/results/e4_scale16.json`, both produced by the hardware submission path of `FINAL_PYTHON/fez_preservation_test.py`. We state plainly what is *not* archived: the per-job runtime metadata — Qiskit Runtime job identifiers, per-circuit shot counts, and the backend calibration timestamps captured at submission — was not retained when these runs were made, and cannot be reconstructed from the surviving artifacts. Table X is therefore reproducible to the level of the archived per-session aggregates, but *not* from raw job records; a reader cannot re-derive an individual circuit’s shot count or the calibration state of the device at submission from this deposit. The `v1.0` release commit hash and the pinned dependency environment `environment.yml` are archived in the DOI-pinned Zenodo deposit (DOI 10.5281/zenodo.19214449), so the aggregate values are traceable to a fixed code revision, and the hardware numbers should be read as session-level measurements of a drifting device rather than as individually re-runnable circuits.

Code availability

The optimization engine, Kraus contraction routines, and experiment drivers are released as open-source Python under the TOPAZ repository (DOI 10.5281/zenodo.19214449, tag `v1.0`).

A1. A noise-model defect in an archived code path

While auditing the provenance of Table IV we located the pipeline that generated it, `run_topology_pruning.py`, in the repository history, and found that it imports its simulator from `V5 module/topaz/harness.py`. In that file, `build_reduced_sim` constructed its two-qubit gate errors as

$$\text{QuantumError}([\text{Kraus}([\text{K0}, \text{K1}, \text{K2}, \text{K3}])]), \quad (\text{A11})$$

with `K0–K3` being 2×2 operators, and registered the result on an instruction spanning *two* qubits. The operator dimension and the instruction arity disagree, and current `qiskit-aer` rejects the construction outright with `NoiseError: Invalid type of input`. The call was wrapped in a bare `except: pass`, so the rejection was swallowed once per edge of the coupling path and the resulting `NoiseModel` contained no

gate errors at all. The density-matrix simulator used for every density-matrix fidelity in that code path therefore returned the noiseless statevector fidelity.

The failure is silent in the most damaging sense: it produces numbers of plausible magnitude rather than an error. We detect it with the elementary consistency check that a noisy fidelity must be strictly below the ideal one,

$$|F_{\text{DM}}^{\text{noisy}} - F_{\text{nl}}| < 10^{-6}, \quad (\text{A12})$$

which held to machine precision (1.9×10^{-15}) for the defective build and fails cleanly for the repaired one (3.3×10^{-2} at the analytic initialisation). The same file already contained the correct construction in a sibling code path, `FINAL_PYTHON/fez_preservation_test.py`, which additionally raises if the assembled noise model has no quantum errors. That path, and therefore the ablation of Table I and the multi-seed ensembles of §6, were never affected; we verified this independently by re-running the nine ablation conditions at ten seeds each and recovering every archived DM and two-qubit-gate value to the reported precision in eight of the nine conditions (Appendix A2).

The defect is a compatibility failure, not a modelling choice: under the toolchain we could not reconstruct, Eq. (A11) may have been accepted, in which case the archived values were noise-aware and correct. We cannot settle that from the repository, because the archived run recorded neither a package lockfile nor its resolved versions, and the `environment.yml` cited in our Data availability statement was not present. We therefore report the defect, retain the archived numbers unchanged, and mark the condition under which they were produced as unestablished. We note that this is precisely the failure mode that a version range rather than a lockfile invites, and we have replaced the range with the resolved versions in Data availability.

A2. Reproduction of the ablation

Table I was regenerated end to end with the recovered producer `run_ablation_battery.py` on Python 3.13 with `qiskit` 2.4.2, `qiskit-aer` 0.17.2 and `qiskit-ibm-runtime` 0.47.0, using the noise model of `src/topaz/harness.py`. Mean density-matrix fidelity and mean two-qubit gate count over ten seeds agree with the archived artifact in eight of the nine conditions, the exception being `cobyla` as noted below:

condition	F_{DM} (archived)	F_{DM} (re-run)	N_{2Q} (archived)	N_{2Q} (re-run)
<code>full_topaz</code>	0.7365	0.7366	12.6	12.6
<code>no_polar</code>	0.3336	0.3337	16.7	16.7
<code>product_only_pte</code>	0.2786	0.2790	16.7	16.7
<code>random_pte</code>	0.0561	0.0560	21.0	21.0
<code>single_mma</code>	0.6226	0.6225	19.9	19.9
<code>adam</code>	0.5136	0.5132	16.5	16.5
<code>cobyla</code>	0.7286	0.7360	15.6	14.9
<code>hea</code>	0.3294	0.3294	28.0	28.0
<code>exact_prep</code>	0.9239	0.9238	21.0	21.0

`cobyla` is the only condition that does not agree, and it differs in both columns ($0.7286 \rightarrow 0.7360$ and $15.6 \rightarrow 14.9$ gates):

it is a derivative-free baseline whose SciPy implementation is not bit-reproducible across versions, and it is not the source of any claim in this paper. Compiled circuit depth is deliberately excluded from this comparison, because it is a property of the transpiler pass rather than of the ansatz and is not bit-stable under a change of Qiskit version; the depth column of Table I is therefore not claimed to reproduce. The dfe column stored in the ablation artifact is identically zero for all ninety records and is therefore not a measurement; it appears in no table here.

Additional information

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TABLE VII Comparison on FakeFcz (quietest path, 4000 shots). LPTE DM is the best seed of the 10-seed ablation ensemble (10-seed mean 0.737 ± 0.066 , Table I). Exact preparation retains a higher DM fidelity (0.924 ± 0.003 at the same 21-gate, `opt_level=3` form; 0.8501 at the 42-gate `opt_level=1` form). The LPTE circuit is *not* pruned at $\tau_k = 0.05$ (§6.6.6), so its advantage over exact preparation is not a depth reduction: both forms use 21 two-qubit gates here, and the difference is the optimization landscape rather than the compiled topology. The mitigation columns in this table are single-trajectory values; the ensemble measurement is Table XI.

Method	DM	Inv.	CDR	VD	2Q	Depth
Exact prep (raw) [†]	0.8501	0.9331	0.2313	0.9930	42	195
Exact + ZNE [†]	0.9813*	0.9326	—	—	42	195
HEA (Ry+CX, 4 lyr.)	0.5332	0.9420	—	—	28	57
TOPAZ product-PTE	0.6329	1.1556	—	—	8	22
TOPAZ LPTE	0.8260	1.1077	0.9349 [§]	0.9964 [§]	21	97
TOPAZ LPTE + ZNE	0.9743*	0.9331	—	—	21	97

[†]Exact-preparation gate counts are reported as-transpiled at `opt_level=1` (the level used when these numbers were produced); the same target collapses to 21 two-qubit gates at depth 82 under `opt_level=3`, the configuration used in the exact-prep robustness footnote of Table II. All other rows are transpiled at `opt_level=3` (§3). *ZNE DM is extrapolated to $\lambda = 0$, not measured. CDR and VD are single-method results applied to the LPTE circuit; they were not run stacked with ZNE. All mitigation methods (ZNE, CDR, VD) are implemented in-house and run against our own density-matrix simulations.

The Exact-prep CDR entry (0.2313) should be read with care: CDR learns a linear model by regression on Clifford circuits sampled near the target, and at the depth-195 exact-preparation circuit that learned response leaves the linear-response regime CDR assumes, so its CDR estimate is unreliable; the shallow depth-97 LPTE circuit is what keeps CDR closer to its validity regime. [§]The LPTE DM entry (0.8260) is the maximum of the 10-seed ablation ensemble (seed 9) and the HEA entry (0.5332) is that ensemble’s HEA maximum (seed 2), both reproduced from the same archived per-seed artifact as Table I. The LPTE CDR and VD entries (0.9349, 0.9964) are *not*: they are best-seed values from a pipeline we have not recovered, and they are not reproduced by the complete 10-seed ensemble of Table XI, which reports CDR at 0.5248 ± 0.1486 and VD at 0.5392 ± 0.1542 on the echo. Table XI supersedes these two entries. Note also that this row’s 21 two-qubit gates are the unpruned chain, consistent with §6.6.6.

Consequently the cost columns of this row and its DM entry describe two different circuits: the archived per-seed artifact records seed 9—the seed that attains 0.8260—as 10 two-qubit gates at depth 24 after pruning, whereas the 21 and 97 printed here are the unpruned chain’s. The row is retained in this form so that the 21-gate figure matches the unpruned chain used throughout §6.6.4 and the hardware comparison below, but the 0.8260 should not be read as the fidelity of the 21-gate circuit. The DM column of this table, and the exact-preparation, product-PTE and ZNE rows generally, are single-trajectory values whose per-seed artifacts we have not recovered; we retain them only to preserve within-table comparability and flag them as such. The **Inv.** column has been recomputed in this revision from `data/noise_transfer/cross_eval.npy` under Eq. (7); the previously reported values in that column did not reproduce. **The Inv. values are indexed by noise model, not by method:** the six entries 0.9331, 0.9326, 0.9420, 1.1556, 1.1077, 0.9331 correspond, in order, to `nominal`, `low_noise`, `high_depol`, `amp_damp`, `phase_damp` and `high_readout` of the transfer matrix, and they are reproduced here in a row order that happens to match the length of the method list. They describe how well a parameter set transfers *out of* each noise model, and they are not robustness scores for the methods named in the first column. The per-noise-model reading, including why two entries exceed 1, is given in the discussion of Fig. 5. We retain the column in place, unchanged in value, so that the original record stays legible, and flag the indexing explicitly rather than silently relocating the numbers.

TABLE VIII τ_κ sensitivity on the 8-qubit benchmark (seeds 0 and 3; results/tau_kappa_sweep_20260926_000726.json). Retained is the number of entangling pairs surviving the threshold out of seven. DM is the noise-aware density-matrix fidelity. This is a *post-hoc* scan: the optimization is run once per seed and the resulting pair intensities are then re-thresholded and the fidelity re-evaluated, which is why the recorded κ vectors are bit-identical across all six rows of a given seed. The fidelity is correspondingly noisy and need not decrease monotonically—at $\tau_\kappa = 0.15$ seed 3 scores higher (0.3685) than at 0.10 (0.0507), because the surviving subset happens to interfere less.

τ_κ	Retained	2Q gates	Depth	DM (seed 0 / seed 3)	
0	7 / 7	21 / 19	79 / 75	0.3485 / 0.5135	Depth is 0 when the
0.01	7 / 7	21 / 19	79 / 75	0.3485 / 0.5135	
0.05	7 / 7	21 / 19	79 / 75	0.3485 / 0.5135	
0.10	6 / 5	18 / 15	68 / 34	0.2912 / 0.0507	
0.15	2 / 3	6 / 9	16 / 22	0.1447 / 0.3685	
0.20	1 / 0	3 / 0	16 / 0	0.1135 / 0.1122	

last pair is removed, since the transpiled circuit is then empty.

TABLE IX Independent ten-seed reproduction (tmp_final10.json). Per-seed final DM fidelity, gate count, depth, and active pairs. The archived ensemble of §7 reports a mean of 0.737 ± 0.066 for the same protocol; the two are reported side by side and not averaged, because the archived producing script has not been recovered and the two numbers are not known to come from identical code paths.

Seed	DM fidelity	2Q gates	Depth	Active pairs
0	0.7770	20	77	7
1	0.6160	18	69	7
2	0.6628	16	62	6
3	0.4761	21	81	7
4	0.5345	18	72	7
5	0.8502	19	69	7
6	0.6903	16	59	7
7	0.8426	20	78	7
8	0.5955	21	86	7
9	0.7885	20	82	7
mean	0.6834	18.9	73.5	6.9
std	0.1295	1.7	8.4	0.3
baseline	0.1425	—	—	—

TABLE X Live `ibm_fez` results, three same-day replications on the quietest path (2026-09-07, path [140, . . . , 147], `opt_level` = 3, $n = 4000$ shots per measurement). Measurement SEMs are binomial ($\sqrt{F(1-F)/n}$); the TOPAZ echo is reported as mean $\pm$ standard deviation across the three sessions (individual values in §6.6.8). Computed (noiseless) entries are exact statevector/density-matrix evaluations of the submitted circuit and carry no shot noise.

Configuration	Type	Fidelity
Product-Trotter, untuned random params (two-way echo)	measured	0.0478 ± 0.0034
Product-Trotter, MMA-optimised (two-way echo)	measured	0.1008 ± 0.0048
TOPAZ LPTE (two-way echo, mean of 3 sessions)	measured	0.2883 ± 0.0394
TOPAZ LPTE (two-way echo, individual sessions)	measured	$0.2443 \pm 0.0068, 0.3008 \pm 0.0073, 0.3200 \pm 0.0074$
ZNE on TOPAZ echo (per session)	measured	0.2058, 0.3075, 0.3968
TPNM on TOPAZ echo (per session)	measured	0.2919, 0.3159, 0.3200
TOPAZ LPTE (one-way)	noiseless	0.5662
Product-Trotter, MMA-optimised (one-way)	noiseless	0.1768
Product-Trotter, untuned random (one-way)	noiseless	0.0758

TABLE XI 8-qubit mitigation on FakeFez over a 10-seed ensemble, reporting mean $\pm$ standard deviation (complete, 10/10 seeds; python/0-FINAL/results/lpte10_optionA/). DM: density-matrix fidelity under the target noise model. Echo: inversion-test fidelity (target-echo), with the noiseless ceiling for the same circuit given for reference.

Configuration	DM Fidelity	Echo Fidelity	Noiseless echo
Random-init baseline ($n=5$)	0.2497	—	—
TOPAZ LPTE alone (no post-processing)	0.5146 $\pm$ 0.1469	0.4702 $\pm$ 0.1366	0.5392 $\pm$ 0.1542
LPTE + ZNE	0.5146 $\pm$ 0.1469	0.4551 $\pm$ 0.1431	—
LPTE + CDR	—	0.5248 $\pm$ 0.1486	—
LPTE + TPNM	—	0.4899 $\pm$ 0.1407	—
LPTE + VD	—	0.5392 $\pm$ 0.1542	0.5392 $\pm$ 0.1542
TOPAZ LPTE + VD (our recommended stack)	—	0.5392 $\pm$ 0.1542	0.5392 $\pm$ 0.1542

Mean $\pm$ s.d. (ddof = 1) over all 10 seeds; no seed is excluded. Each method is applied individually to the LPTE circuit; a combined ZNE+CDR+VD stack was not run, so the rows are not additive. The two observable columns are reported separately because the methods divide cleanly between them: ZNE is the only method evaluated on the DM fidelity, and CDR, TPNM and VD on the echo. ZNE extrapolates to $\lambda = 0$, which is precisely the noiseless operator, so *its DM entry is identically equal to the unmitigated one in all 10 seeds*—it is a null operation on that observable by construction, not a measured null effect. On the echo, by contrast, ZNE *does* reduce fidelity (0.4702 $\rightarrow$ 0.4551, -0.0151), and we report that as a negative result rather than omitting it. VD is computed as $(\rho^2)_{00}/\text{Tr}(\rho^2)$, capped at the noiseless echo, so its agreement with that ceiling is a property of the metric’s definition rather than independent evidence of recovery: in 8 of 10 seeds the capped value equals the ceiling exactly and in the remaining 2 it falls short by $< 7 \times 10^{-5}$, and it never exceeds it. The defensible claim is therefore that VD removes the entire residual noise penalty on the echo observable, not that it recovers beyond the ideal. TPNM is projected physical-map noise mitigation. The baseline row is the unoptimized random-initialization DM fidelity from the 5-seed artifact data/multi_seed/fez_results.npy (0.2497 $\pm$ 0.1417, ddof = 0); no echo-side baseline was recorded for that ensemble. An earlier draft quoted 0.0677 here, which is the *seed-0 value only* from that artifact rather than a mean, and we correct it in this revision. The 10-seed mitigated rows and the 5-seed baseline are different ensembles and are not to be differenced against one another.

TABLE XII Measured dimension dim I of the Lie closure of the $3(N - 1)$ per-pair generators of the LPTE chain, compared with the maximum $\dim \mathfrak{su}(2^N) = 4^N - 1$. Rows $N \leq 8$ are exact (enumeration completed).

N	dim I	dim $\mathfrak{su}(2^N)$	dim I / dim $\mathfrak{su}$
4	60	255	0.2353
5	255	1023	0.2493
6	1020	4095	0.2491
7	4095	16383	0.2500
8	16380	65535	0.2499

TABLE XIII Positioning TOPAZ against prior approaches that adapt circuit structure. “Continuous”: edge presence is relaxed to a real-valued intensity during optimization. “Prune stage”: unused operators/edges are removed at some point in the pipeline.

Method	Selection	Continuous	Prune stage	Noise-aware
ADAPT-VQE [23]	grow, max-gradient pool	no	no	no
Qubit-ADAPT-VQE [38]	grow, max-gradient Pauli pool	no	no	no
Pruned-ADAPT-VQE [39]	grow + amplitude removal	no	yes (post-hoc)	no
Plateau-operator elimination [40]	grow + operator removal	no	yes (in-pool)	no
TopGen [41]	discrete bottom-up search	no	yes (sub-circuit merge)	no
TOPAZ (this work)	continuous density + polar projection	yes	yes (pre-compile, τ_e)	yes